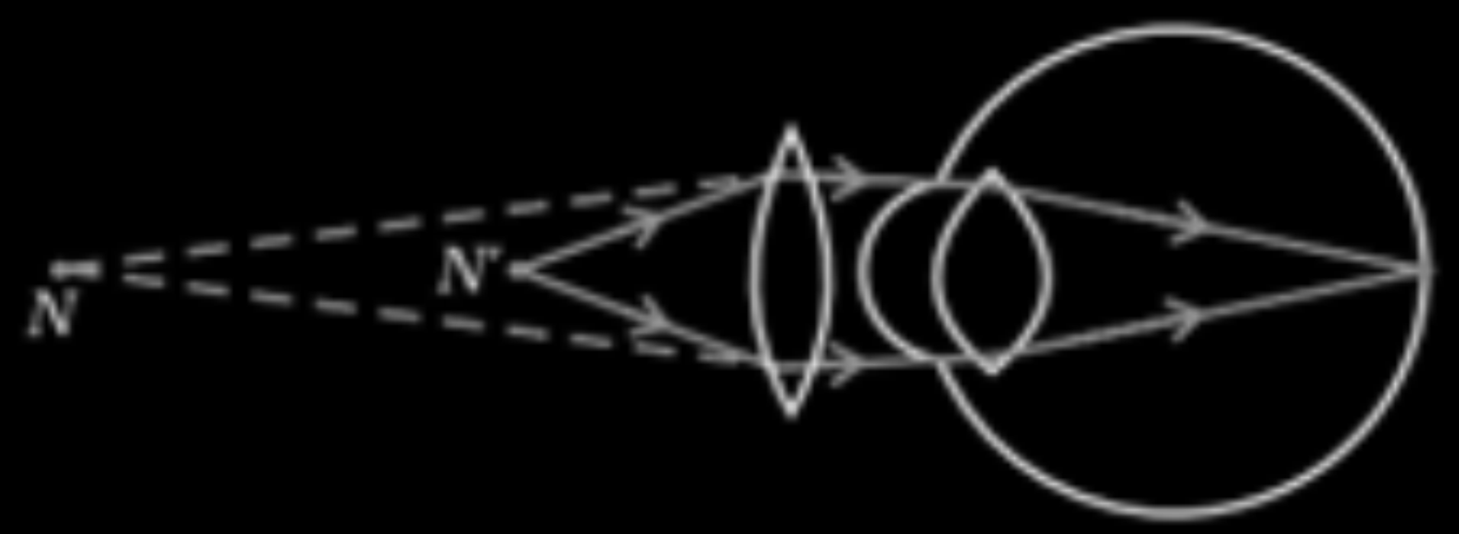
35. A person is unable to read a book placed closer than 1 meter from his eyes. Identify the defect of vision in his eyes. Draw the ray diagrams to show the defect of vision and its correction. **CBSE 26** **3**

35. A person is unable to read a book placed closer than 1 meter from his eyes. Identify the defect of vision in his eyes. Draw the ray diagrams to show the defect of vision and its correction. CBSE 26 **3**

• Hypermetropia/ Far sightedness	**1**
•  *Hypermetropic eye*	**1**
•  *Correction for Hypermetropic eye*	**1**
(deduct ½ mark for not showing the direction of ray of light)	

37. (a) What is hypermetropia ? **3**

(b) Write any one cause of hypermetropia.

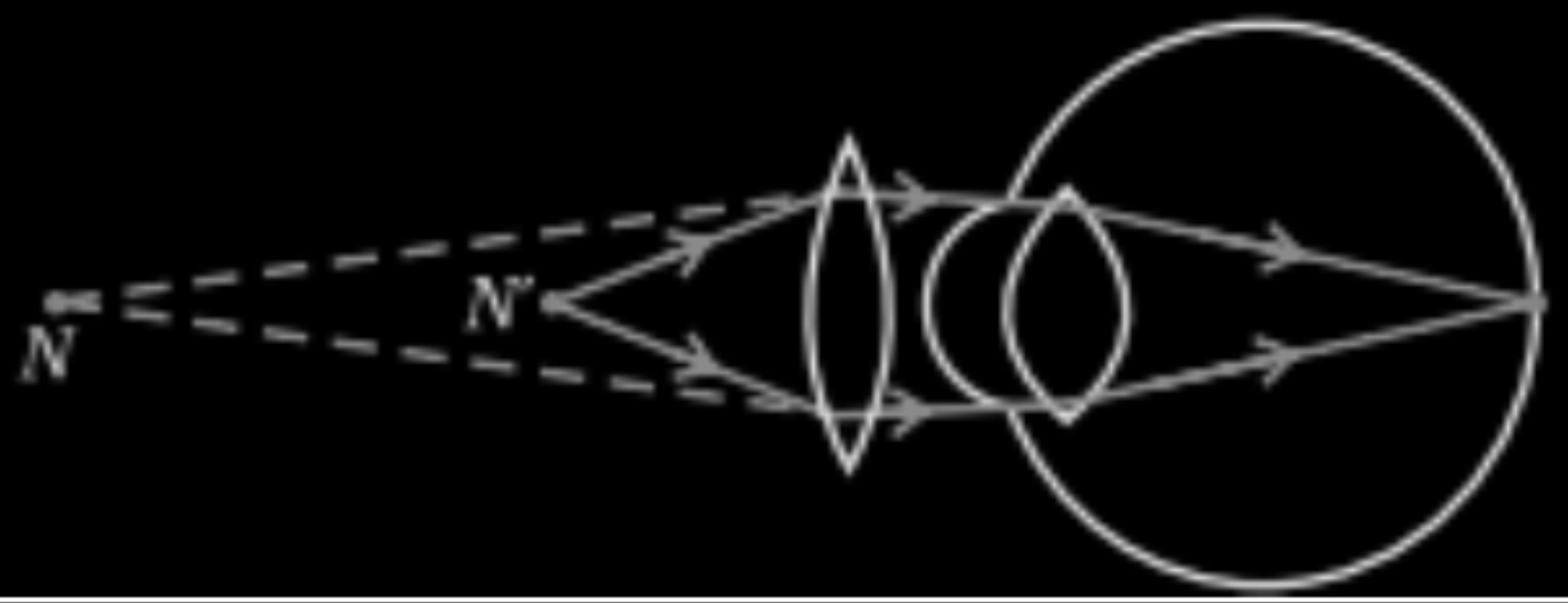
(c) With the help of a suitable ray diagram, explain how hypermetropia is corrected ?

37. (a) What is hypermetropia ? **3**

(b) Write any one cause of hypermetropia.

(c) With the help of a suitable ray diagram, explain how hypermetropia is corrected ?

CBSE 26

(a) An eye defect in which a person can see distant objects clearly but cannot see nearby object clearly.	**1**
(b) The focal length of the eye lens is too long. / The eyeball has become too small.	**1**
(c) 	**1**

(b) Why is the concave lens used as a corrective lens for a myopic eye ? *2*

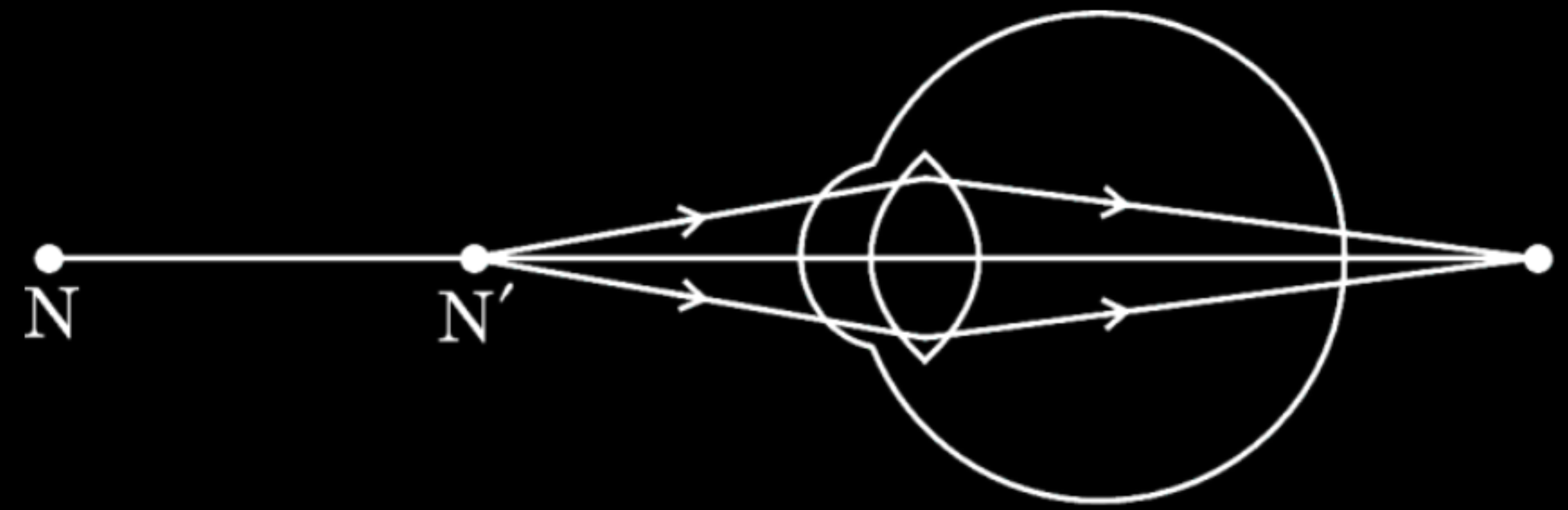
CBSE 26

(b) Why is the concave lens used as a corrective lens for a myopic eye ? *2*

CBSE 26

(b)	In myopic eye image is formed in front of the retina. A concave lens / diverging lens of suitable power will bring the image back on to the retina.	**2**

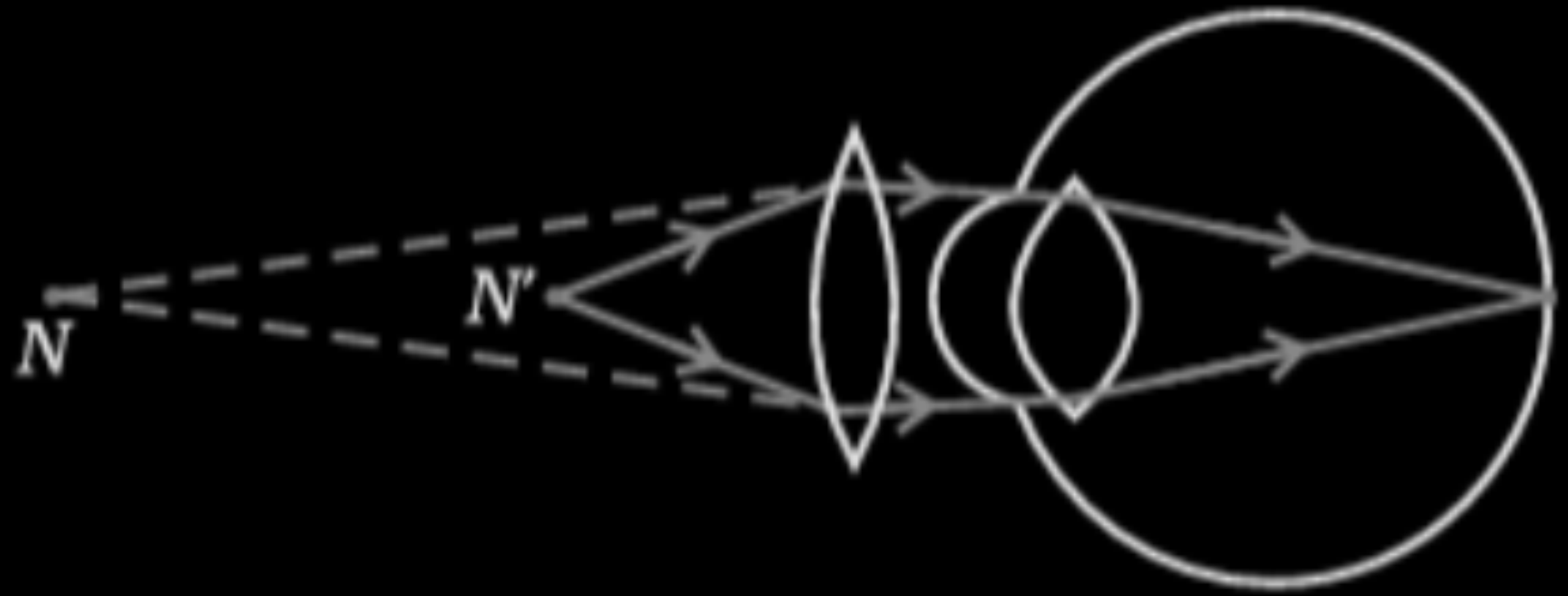
35. Study the given diagram and answer the questions that follow : CBSE 26



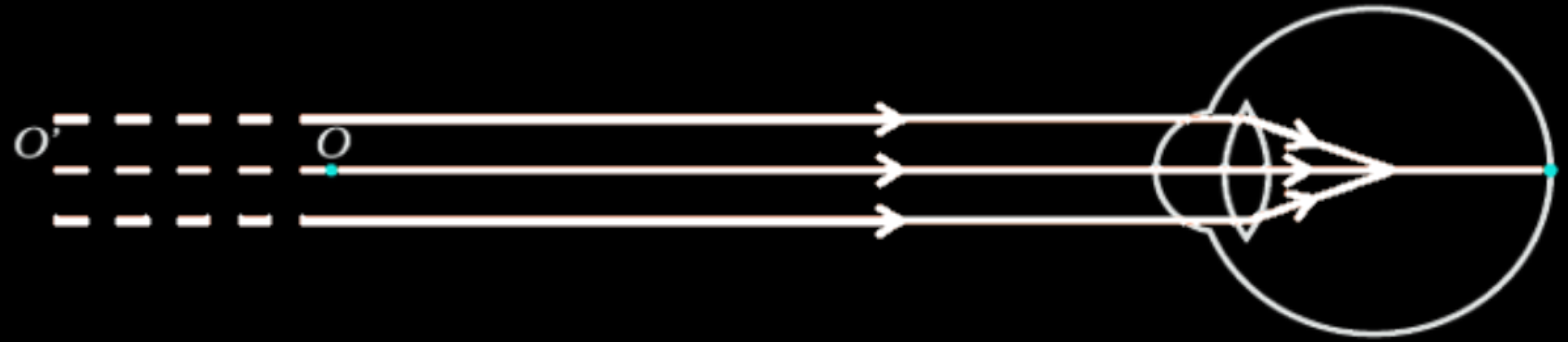
(a) Write the name of the eye defect shown in the diagram. Where is the image formed in this eye defect with respect to the retina of the eye ?

(b) List two causes of this eye defect.

(c) With the help of a diagram, show how this eye defect of vision is corrected. *3*

(a)	
• Hypermetropia/ Far -Sightedness	½
• Behind the retina.	½
(b)	
• Focal length of the eye lens is too long.	½
• The eye ball has become too small.	½
(c)	
	1

37. Study the given diagram and answer the questions that follow : *3*

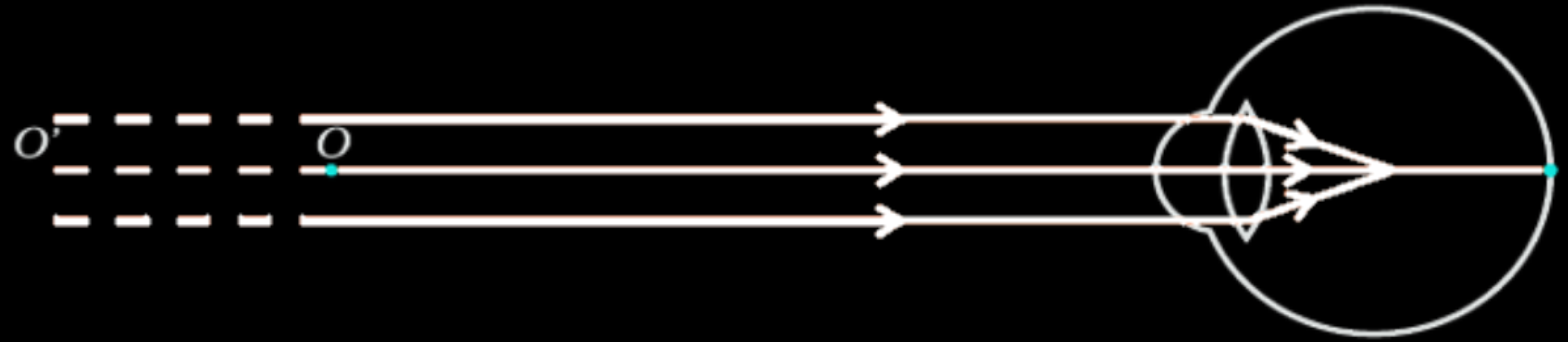


(a) Write the name of the eye defect shown in the diagram. Where is the image formed in this eye defect with respect to the retina of the eye ?

(b) List two causes of this eye defect.

(c) With the help of a ray diagram, show how this defect of vision is corrected. **CBSE 26**

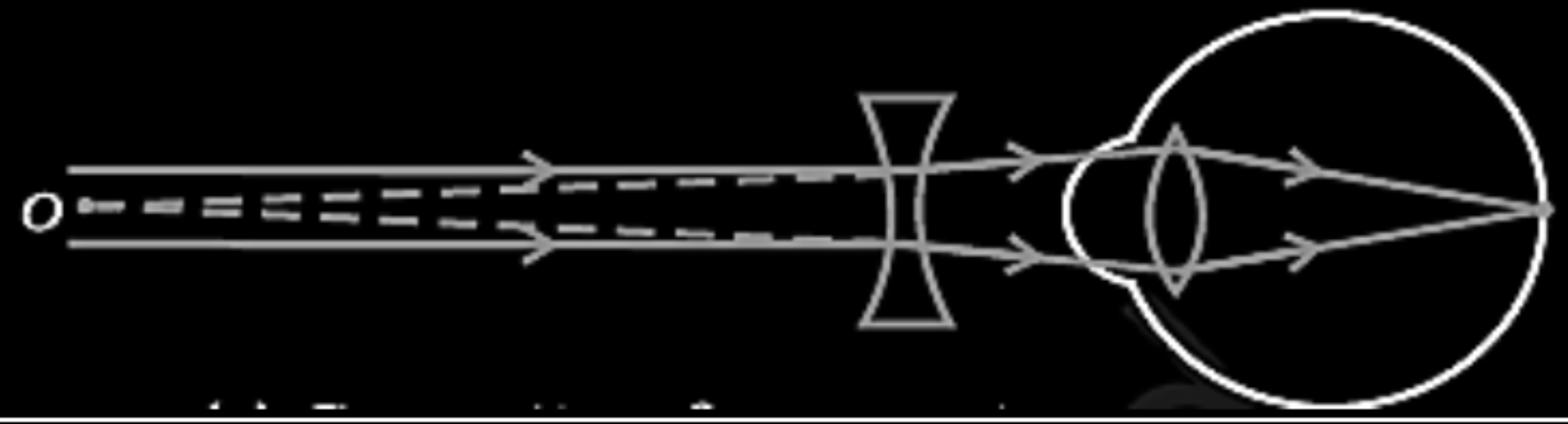
37. Study the given diagram and answer the questions that follow : *3*



(a) Write the name of the eye defect shown in the diagram. Where is the image formed in this eye defect with respect to the retina of the eye ?

(b) List two causes of this eye defect.

(c) With the help of a ray diagram, show how this defect of vision is corrected.

	CBSE 26
(a) • Myopia • The image is formed in front of the retina.	½ ½
(b) • Excessive curvature of the eye lens. • Elongation of the eye ball.	½ ½
(c) 	1

31. The vision defect which arises due to gradual weakening of the ciliary muscles and diminishing flexibility of eye lens is : **1**

(A) Myopia

(B) Hypermetropia

(C) Presbyopia

(D) Myopia and hypermetropia both

31. The vision defect which arises due to gradual weakening of the ciliary muscles and diminishing flexibility of eye lens is : **1**

(A) Myopia (B) Hypermetropia

(C) Presbyopia (D) Myopia and hypermetropia both

CBSE 26

31. The vision defect which arises due to gradual weakening of the ciliary muscles and diminishing flexibility of eye lens is : **1**

(A) Myopia

(B) Hypermetropia

(C) Presbyopia

(D) Myopia and hypermetropia both

CBSE 26

(C) / Presbyopia

31. To restore clear vision in persons whose size of the eye ball has reduced, he/she is suggested to use suitable **1**

(A) Converging lens

(B) Diverging lens

(C) Bifocal lens

(D) Cylindrical lens

CBSE 26

31. To restore clear vision in persons whose size of the eye ball has reduced, he/she is suggested to use suitable **1**

(A) Converging lens (B) Diverging lens

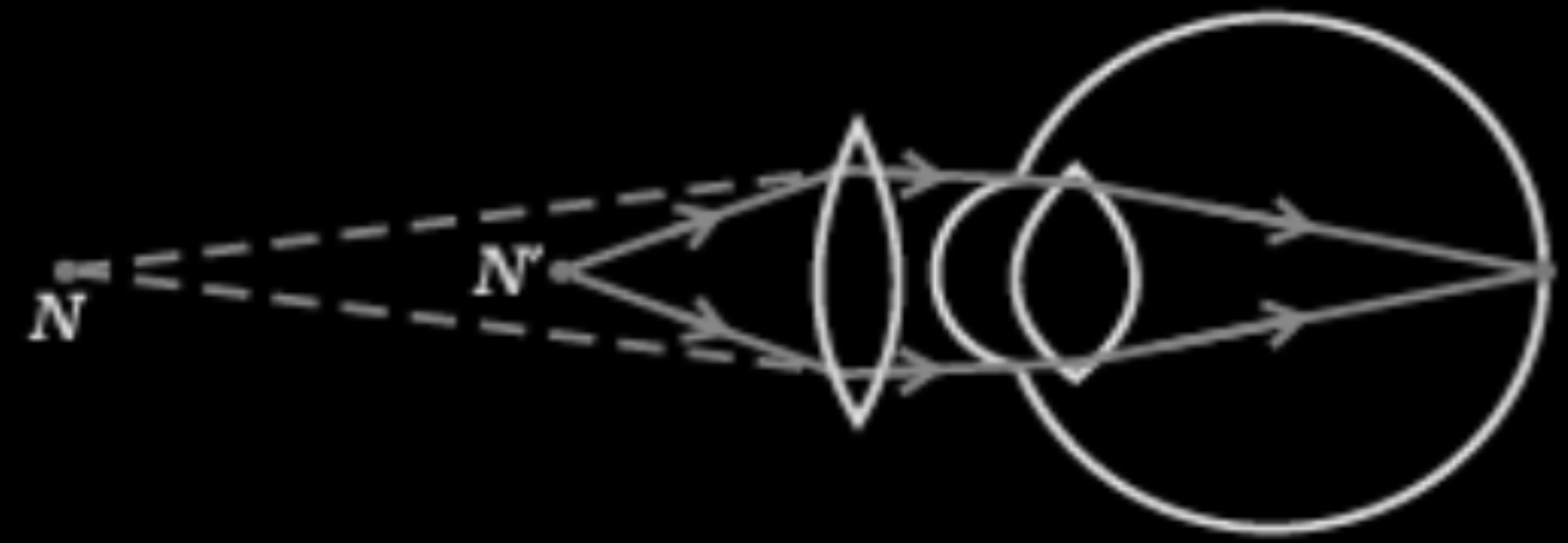
(C) Bifocal lens (D) Cylindrical lens

CBSE 26

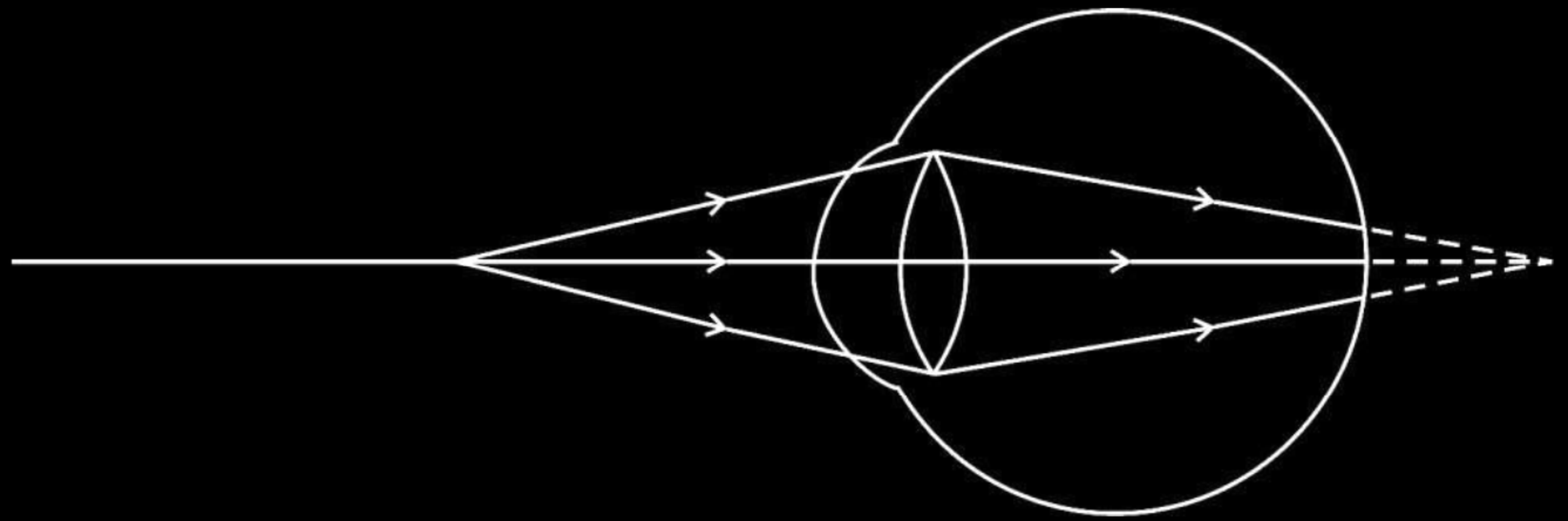
(A) / converging lens

(B) Draw a ray diagram to show the correction of eye defect of an old man who can not see an object placed closer than 1 m from his eye, clearly. **CBSE 26**

(B) Draw a ray diagram to show the correction of eye defect of an old man who can not see an object placed closer than 1 m from his eye, clearly. **CBSE 26**

 (Deduct ½ mark for not showing the direction of light)	**2**

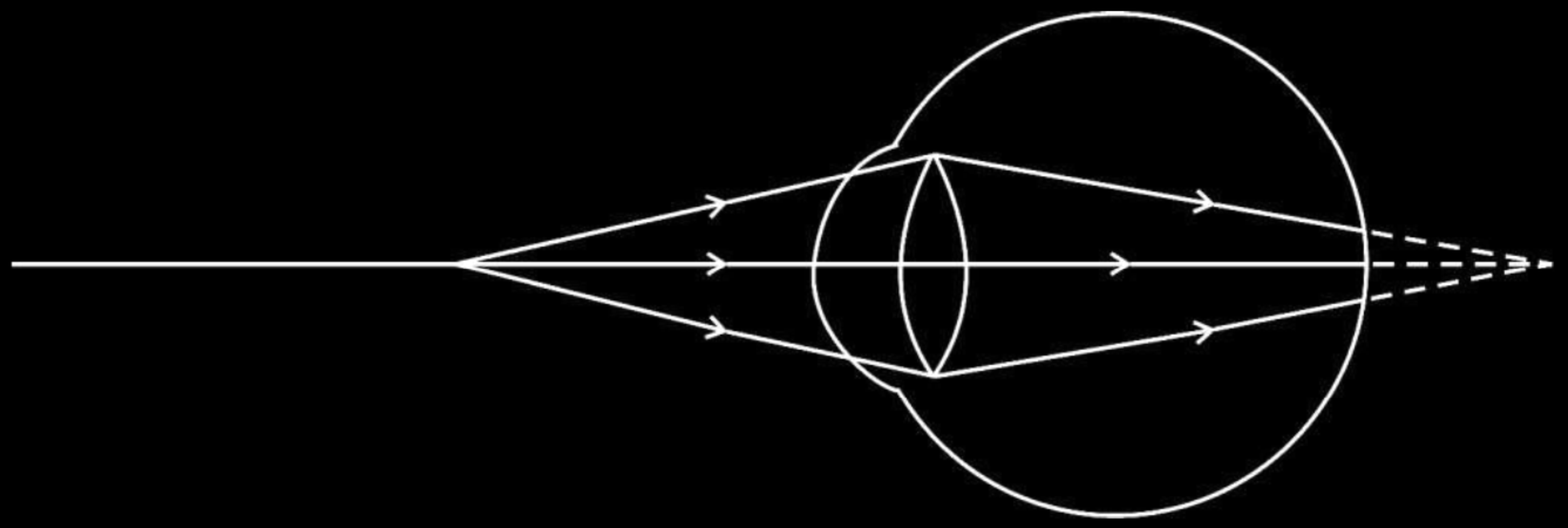
26. Observe the following diagram showing an image formation in an eye :



(a) Identify the defect of vision shown in the figure.

(b) List its two causes and suggest a suitable corrective lens to overcome this defect. *2*

26. Observe the following diagram showing an image formation in an eye :



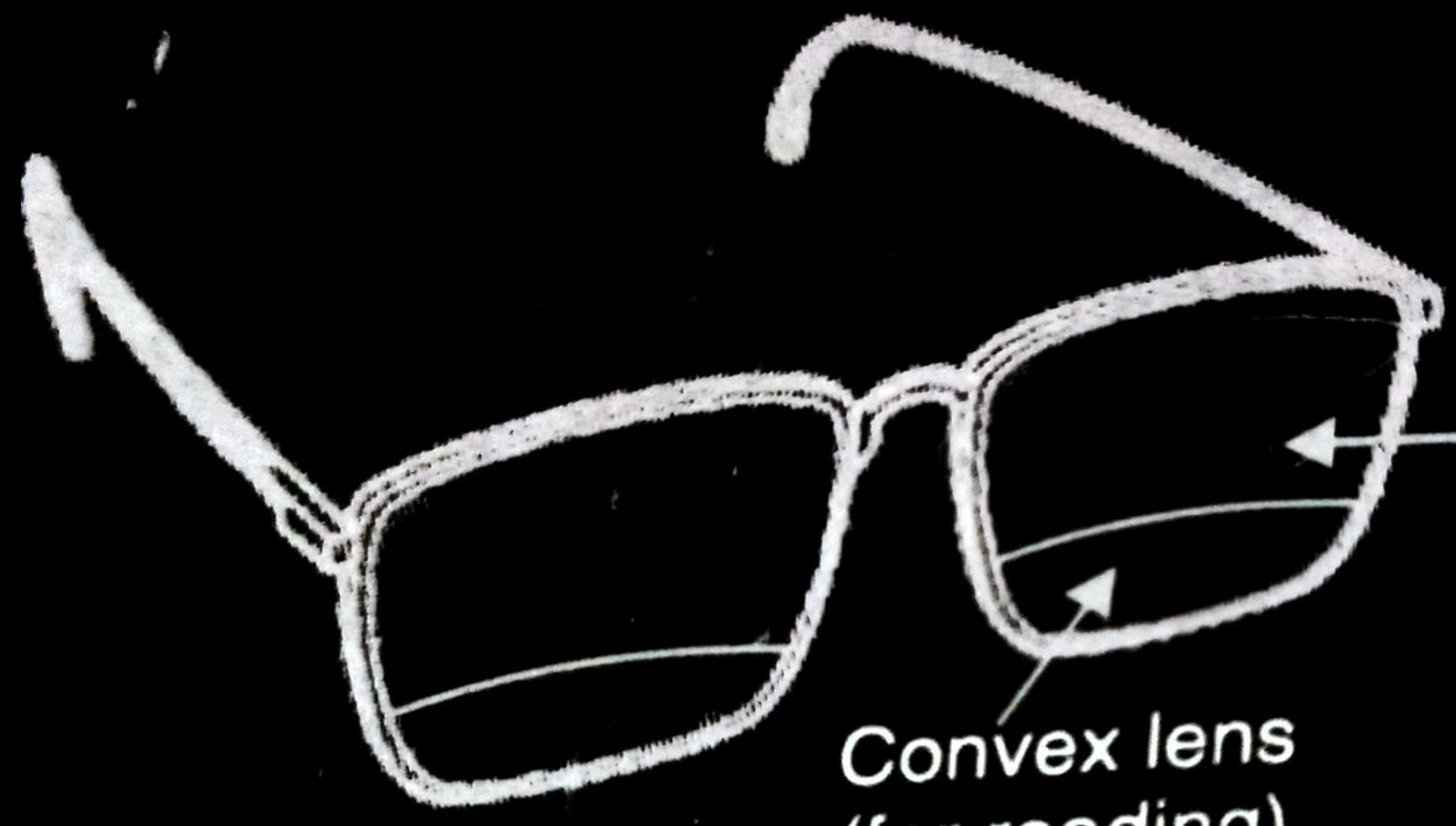
(a) Identify the defect of vision shown in the figure.

(b) List its two causes and suggest a suitable corrective lens to overcome this defect. *2*

(a) Hypermetropia/Far sightedness	½
(b) •Focal length is too long	½
•Size of eyeball is small.	½
•Convex lens/Converging lens	½

29. Draw a labelled diagram of a common type of bifocal lens. State the function of each of its parts. Name the defect for the correction of which it is used and state its main cause. *3*

29. Draw a labelled diagram of a common type of bifocal lens. State the function of each of its parts. Name the defect for the correction of which it is used and state its main cause. *3*

	1
• The concave lens (upper portion of bifocal lens) facilitates distant vision.	½
•The convex lens (lower portion of bifocal lens) facilitates near vision.	½
•Presbyopia	½
- Diminishing flexibility of the eye lens.	
- Gradual weakening of ciliary muscles	½
(any one cause)	

20. *Assertion (A) :* A person suffering from myopia cannot see the distant objects clearly.

Reason (R) : A converging lens is used for the correction of myopic eye as it can form real as well as virtual images of the objects placed in front of it.

20. *Assertion (A) :* A person suffering from myopia cannot see the distant objects clearly.

Reason (R) : A converging lens is used for the correction of myopic eye as it can form real as well as virtual images of the objects placed in front of it.

(c) Assertion (A) is true, but Reason (R) is false.

25. (a) (i) What is the (I) least distance of distinct vision, and (II) far point of a normal human eye ?

(ii) A student is unable to see distinctly a chart hanging on the wall of his classroom. Name the defect of vision he is suffering from and the type of corrective lens he must use in his spectacles. *2*

25. (a) (i) What is the (I) least distance of distinct vision, and (II) far point of a normal human eye ?

(ii) A student is unable to see distinctly a chart hanging on the wall of his classroom. Name the defect of vision he is suffering from and the type of corrective lens he must use in his spectacles. *2*

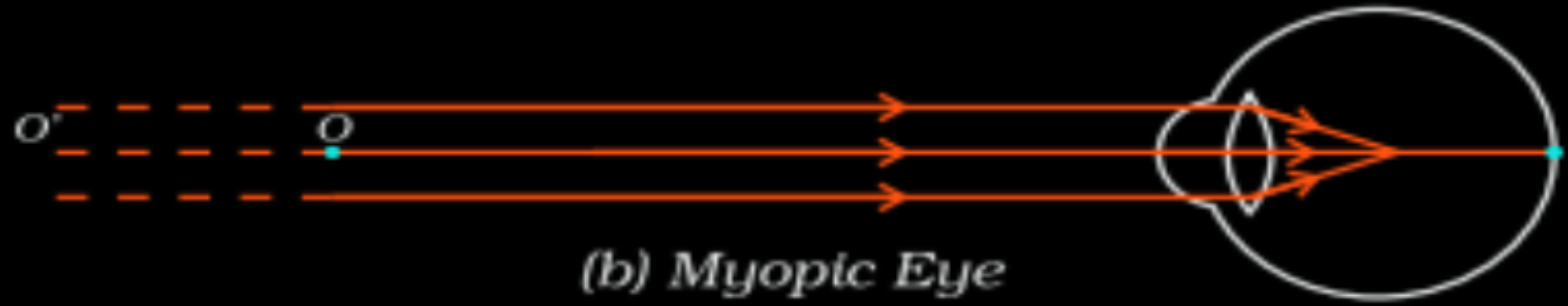
(a)	
(i) (I) 25 cm (II) infinity / ∞	½, ½
(ii) • myopia/near sightedness • concave lens/diverging lens	½, ½

31. A person is unable to see clearly a poster fixed on a distant wall. He however sees it clearly when standing at a distance of about 2 m from the wall.

(a) Draw ray diagram to show the formation of image by his eye lens when he is far away from the wall.

(b) List two possible reasons of this defect of vision.

(c) Draw ray diagram to show the correction of this defect using appropriate lens. **3**

(a)  (b) Myopic Eye	1
(b)	
• Excessive curvature of the eye lens	½
•Elongation of the eyeball	½
(c)  (c) Correction for myopia	1

28. State reasons for Myopia. With the help of ray diagrams, show the

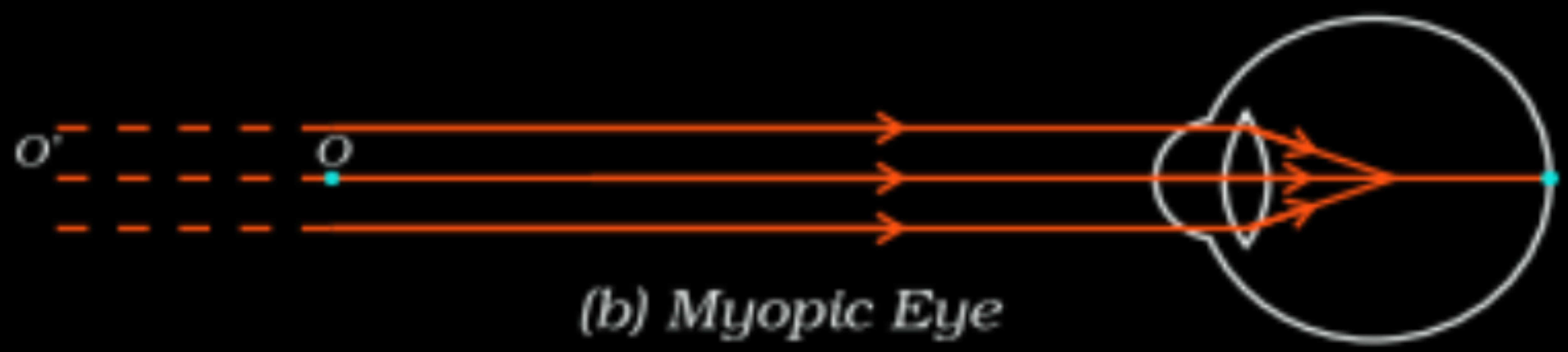
(a) image formation by a myopic eye, and

(b) correction of myopia using an appropriate lens. **3**

28. State reasons for Myopia. With the help of ray diagrams, show the

(a) image formation by a myopic eye, and

(b) correction of myopia using an appropriate lens. **3**

• excessive curvature of the eye lens	½
• elongation of the eyeball	½
(a)  (b) Myopic Eye	1
(b)  (c) Correction for myopia	1

33. A person cannot see distinctly the object placed beyond 5 m from his eyes. Name the defect of vision the person is suffering from. Draw a ray diagram to illustrate this defect. List its two possible causes. Name the lens used for the correction of this defect. **3**

(b) It is observed that the power of an eye to see nearby objects as well as far off objects diminishes with age. **2**

(i) Give reason for the above statement.

(ii) Name the defect that is likely to arise in the eyes in such a condition.

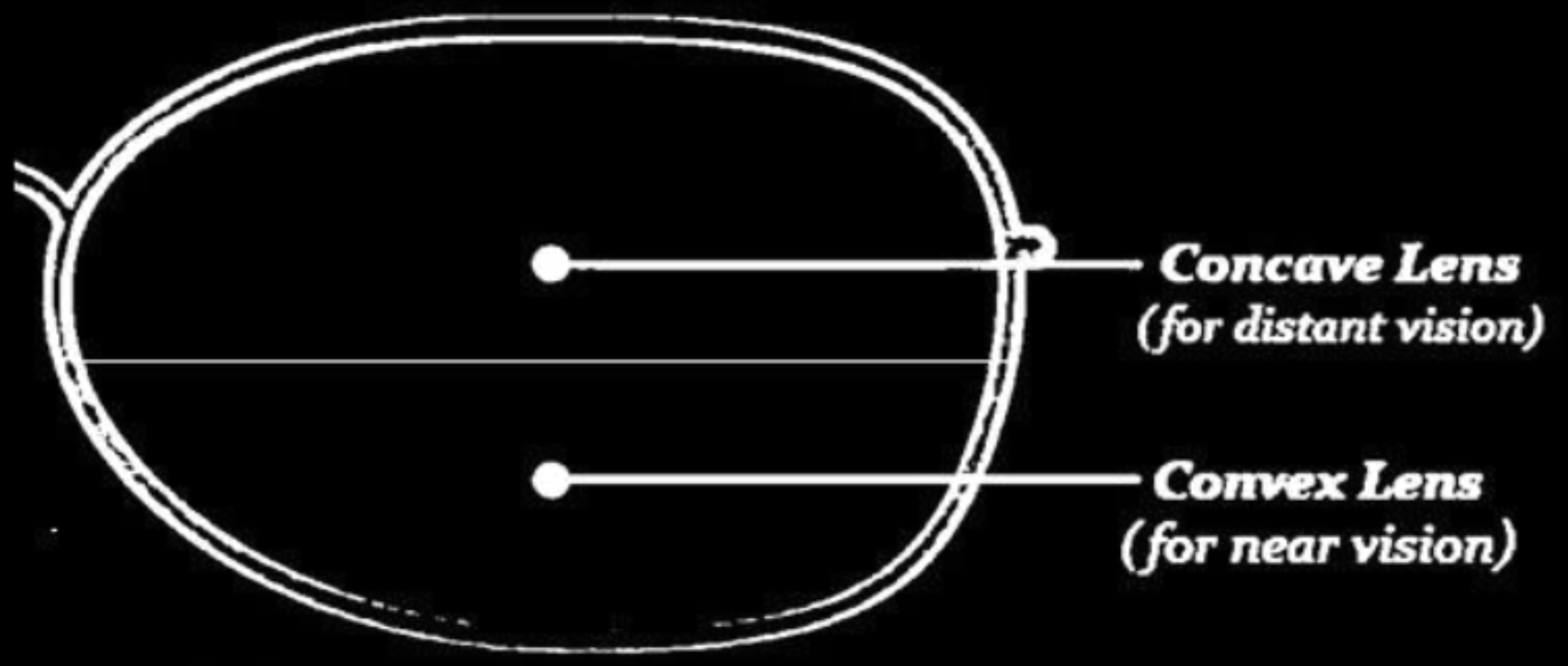
(iii) Draw a labelled ray diagram to show the type of corrective lens used for restoring the vision of such an eye.

(b) It is observed that the power of an eye to see nearby objects as well as far off objects diminishes with age. **2**

(i) Give reason for the above statement.

(ii) Name the defect that is likely to arise in the eyes in such a condition.

(iii) Draw a labelled ray diagram to show the type of corrective lens used for restoring the vision of such an eye.

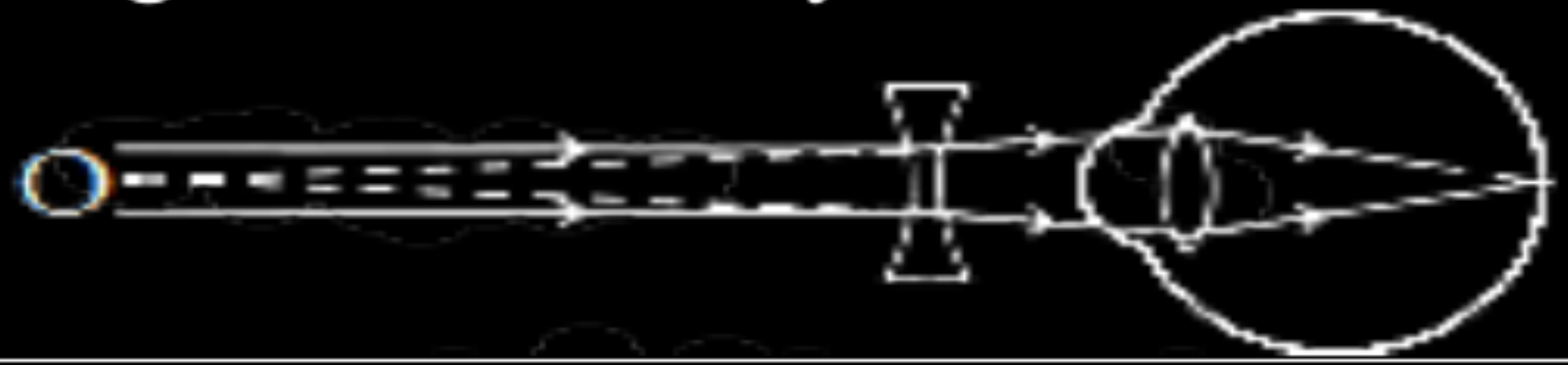
(b) (i) It is due to gradual weakening of the ciliary muscles and diminishing flexibility of the eye lens.	½, ½
(ii) Presbyopia/ Presbyopia + Myopia	½
(iii) Bifocal /Concave + Convex lens/ Diagram	½
 	

29. A person is suffering from an eye defect in which the far point of the eye is nearer than infinity. Identify the defect. List two main causes of this defect. **3**

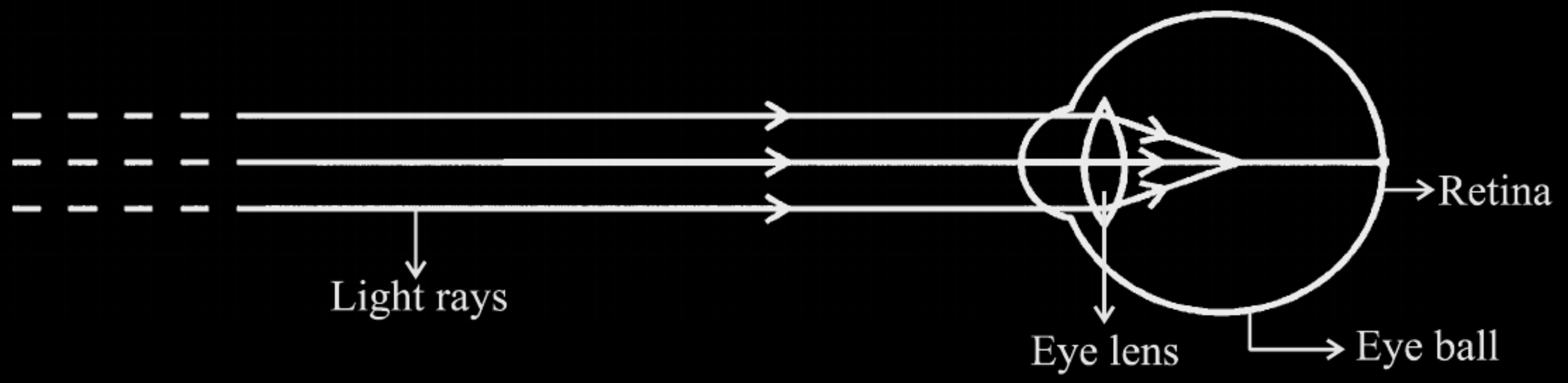
Draw a ray diagram to show how this defect is corrected by using a suitable lens.

29. A person is suffering from an eye defect in which the far point of the eye is nearer than infinity. Identify the defect. List two main causes of this defect. **3**

Draw a ray diagram to show how this defect is corrected by using a suitable lens.

•Defect : Myopia/Short sightedness	1
Two Causes :	
• Excessive curvature of the eye lens	½
• Elongation of the eyeball	½
	1

26. **(A)** Observe the following diagram and answer the questions following it : **2**

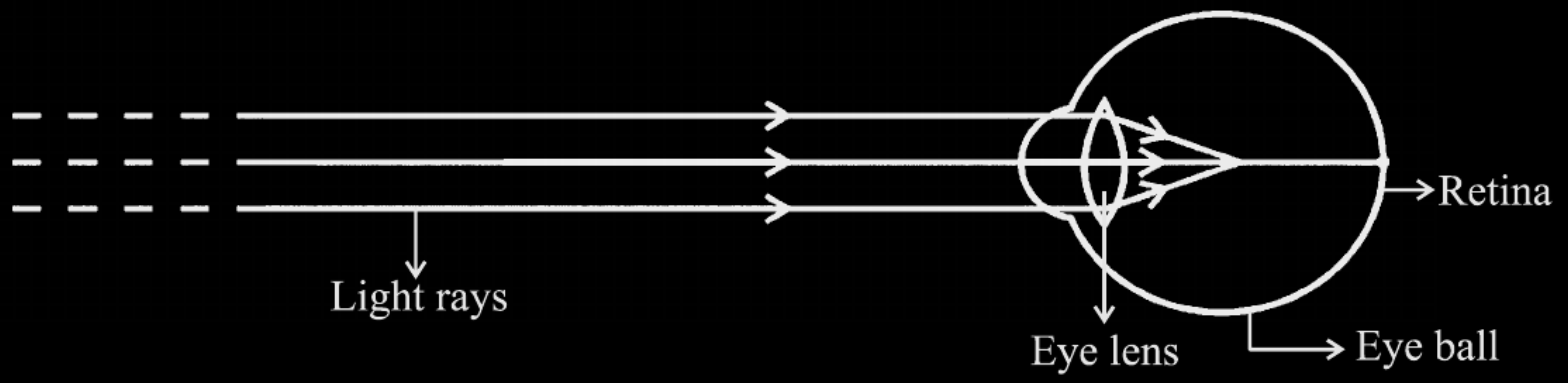


(i) Identify the defect of vision shown.

(ii) List its two causes.

(iii) Name the type of lens used for the correction of this defect.

26. **(A)** Observe the following diagram and answer the questions following it : **2**



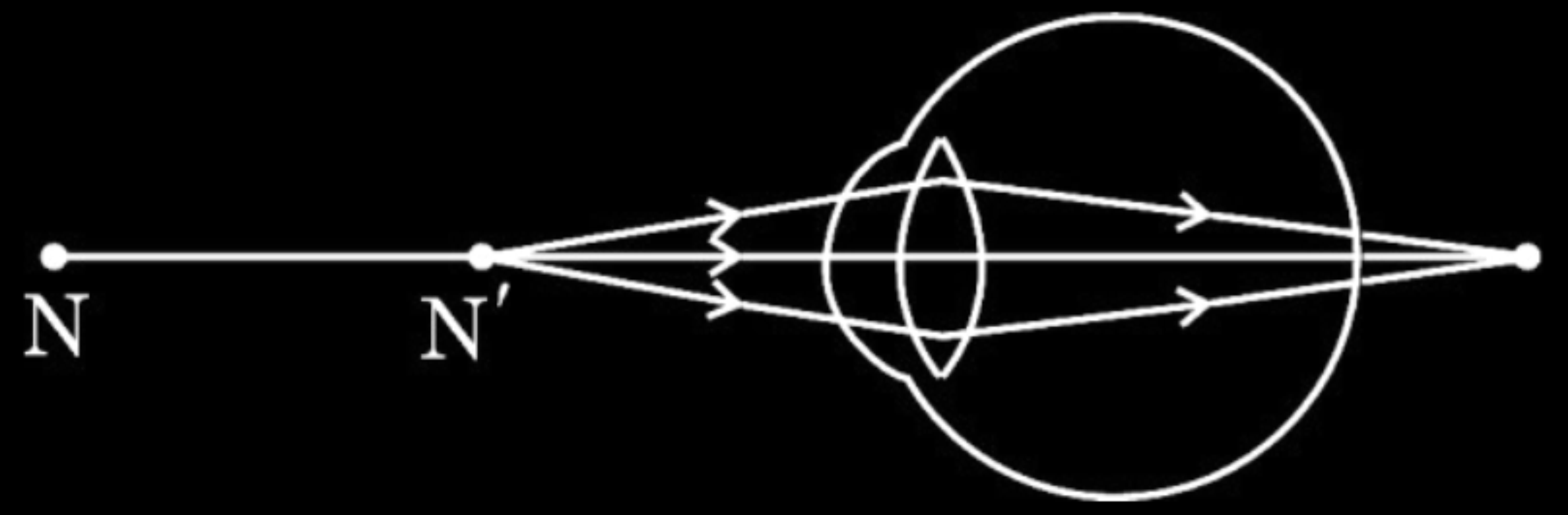
(i) Identify the defect of vision shown.

(ii) List its two causes.

(iii) Name the type of lens used for the correction of this defect.

(A) (i) Myopia / Short Sightedness	½
(ii) • Excessive curvature of eye lens	½
• Elongation of eye ball	½
(iii) Concave lens /Diverging Lens	½

33. (a) Study the diagram given below and answer the questions that follow :



(i) Name the defect of vision depicted in this diagram stating the part of the eye responsible for this condition.

(ii) List two causes of this defect.

(iii) Name the type of lens used to correct this defect and state its role in this case. *3*

(a)	
(i) • Hypermetropia	½
• Ciliary muscles/ eye lens	½
(ii) • Focal length of the eye lens is too long.	½
• Eyeball becomes too small.	½
(iii) Converging lenses/ convex lens	½
They provide the additional focussing power required for forming the image on the retina./ Decrease the focal length of the eye lens	½

25. (a) A student cannot see distinctly the charts hanging in the science laboratory. Name the defect of vision he/she is suffering from. List two causes of this defect. Suggest the type of lenses for the modification of this defect. *2*

25. (a) A student cannot see distinctly the charts hanging in the science laboratory. Name the defect of vision he/she is suffering from. List two causes of this defect. Suggest the type of lenses for the modification of this defect. *2*

(a) • Myopia / Near sightedness	½
Causes: i. Elongation of eyeball	½
ii. Excessive curvature of eye lens	½
• Modification – use of concave lens / diverging lens	½

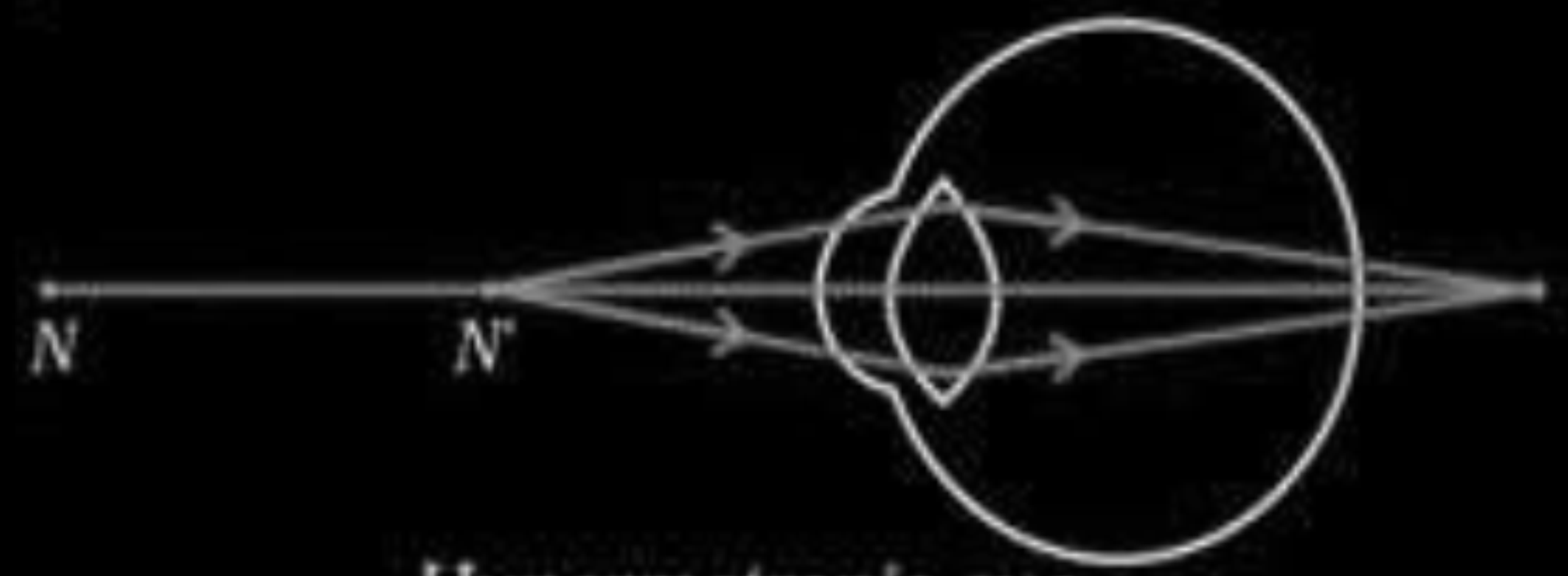
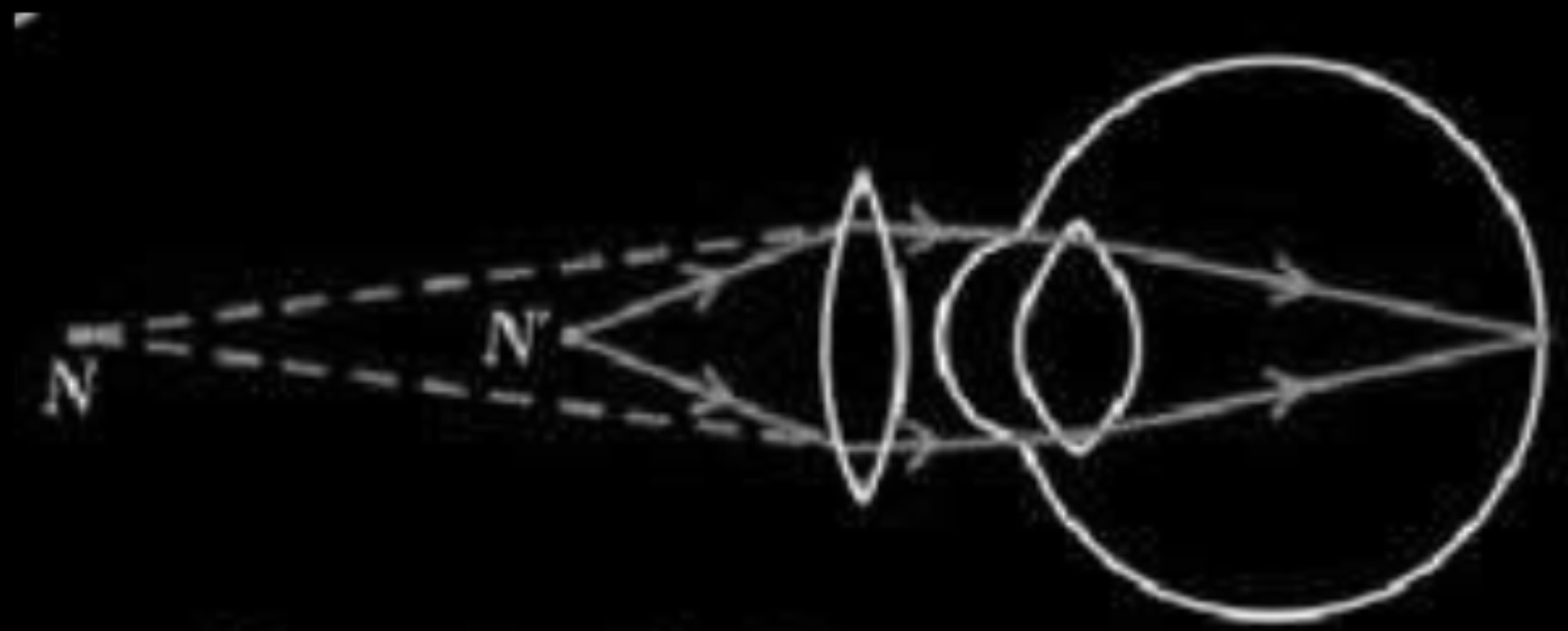
(b) A person cannot see distinctly the objects placed at the least distance of distinct vision for a normal eye. However, he can easily read a newspaper by placing it at a distance of 40 cm from his eyes. Name the defect of vision in this case. List its two causes. Suggest the type of lenses for his spectacles. *2*

(b) A person cannot see distinctly the objects placed at the least distance of distinct vision for a normal eye. However, he can easily read a newspaper by placing it at a distance of 40 cm from his eyes. Name the defect of vision in this case. List its two causes. Suggest the type of lenses for his spectacles. *2*

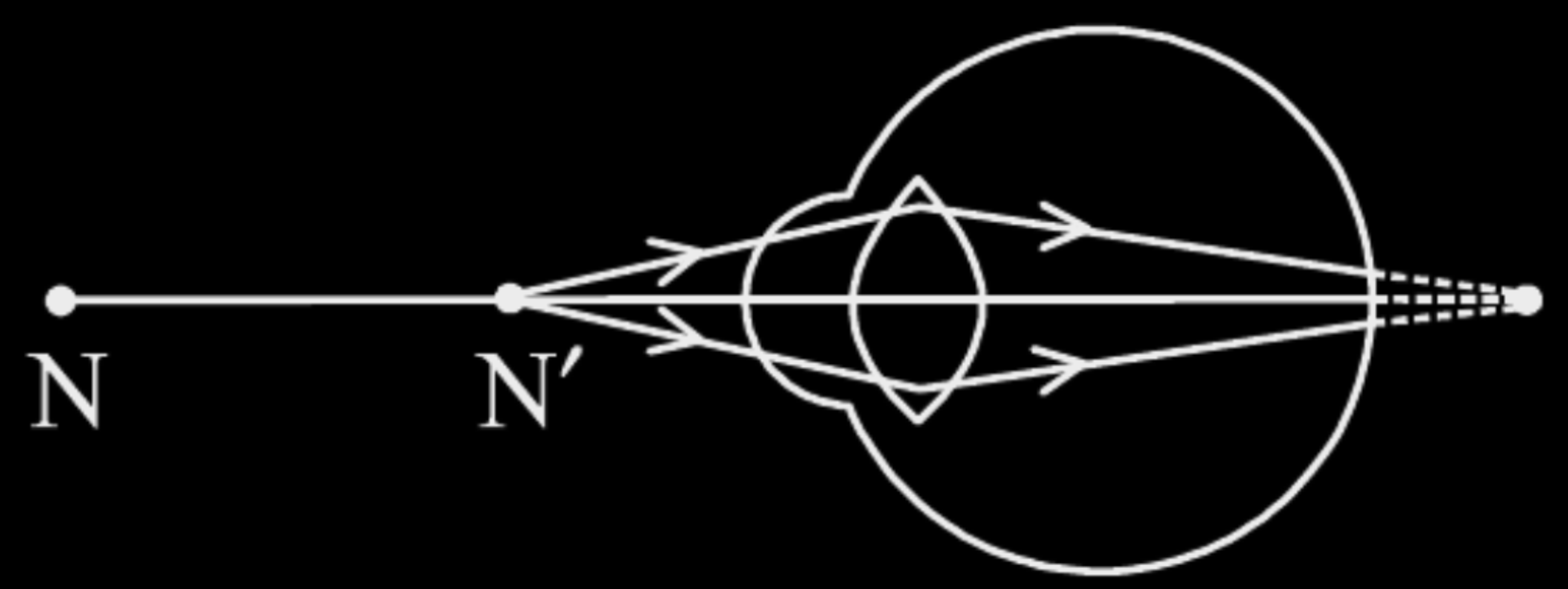
b. • Hypermetropia / Far sightedness	½
Causes: i. Focal length of the eye lens increases	½
ii. Eyeball becomes too small	½
• convex lens / converging lens	½

31. A person keeps the reading material much beyond 25 cm from the eye for comfortable reading. Name the defect of vision he is suffering from. How can it be corrected ? Draw ray diagrams for (i) the defective eye and (ii) its correction. *3*

31. A person keeps the reading material much beyond 25 cm from the eye for comfortable reading. Name the defect of vision he is suffering from. How can it be corrected ? Draw ray diagrams for (i) the defective eye and (ii) its correction. *3*

• Hypermetropia / far sightedness.	½
• By using a convex lens of appropriate power	½
(i)  *Hypermetropic eye*	1
(ii)  *Correction for Hypermetropic eye*	1

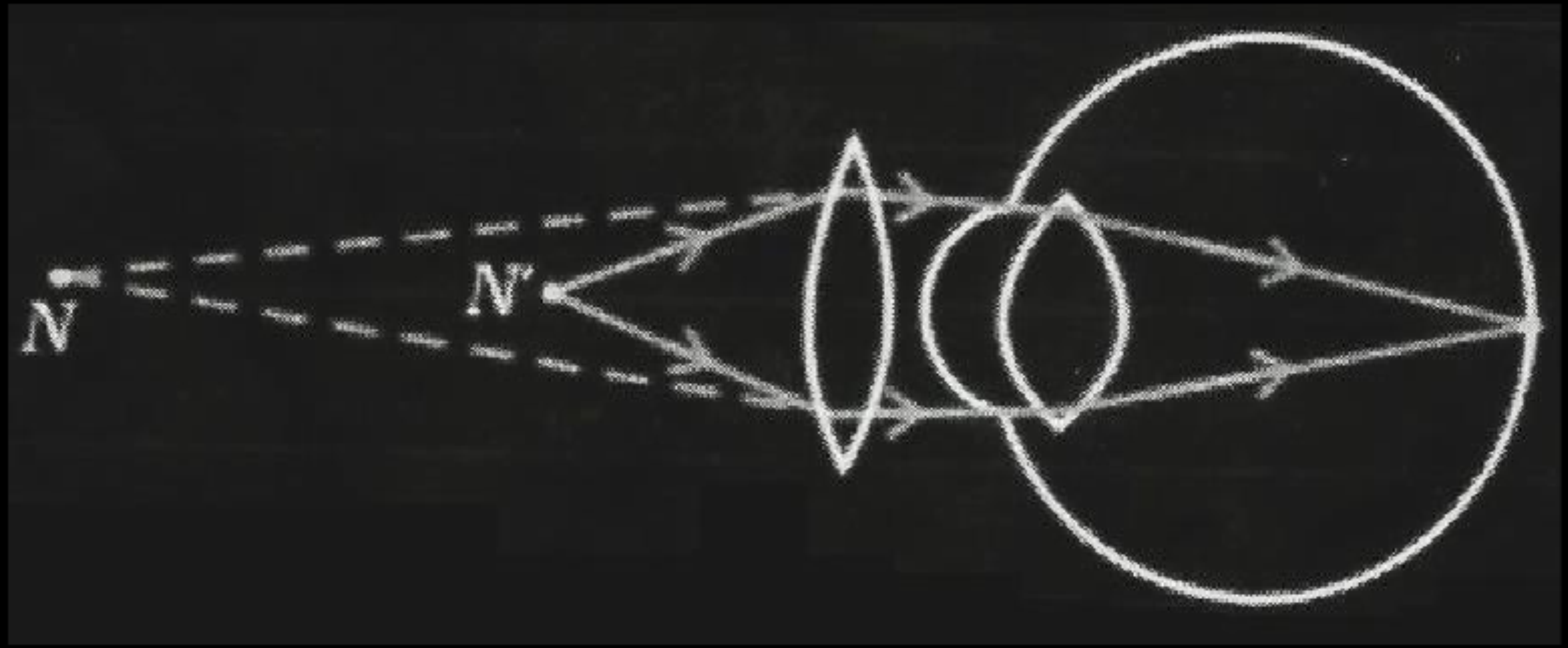
31. Study the diagram given below and answer the questions that follow : **3**



(i) Name the defect of vision represented in the diagram. Give reason for your answer.

(ii) List two causes of this defect.

(iii) With the help of a diagram show how this defect of vision is corrected.

(i) Hypermetropia or Far-sightedness.	½
Reason – Image is formed behind the retina. / Near point for the person is farther away from the normal near point (25 cm)	½
(ii)	
• Focal length of the eye lens is too long.	½
• The eyeball has become too small.	½
(iii) 	1
N = Near point of a hypermetropic eye N'= Near point of a normal eye	

14. In bifocal lenses used for the correction of presbyopia : **1**

(a) the upper portion is of convex lens for the near vision and lower part is of concave lens for the distant vision.

(b) the upper portion is of convex lens for the distant vision and lower part is of concave lens for the near vision.

(c) the upper portion is of concave lens is for the near vision and lower part is of convex lens for the distant vision.

(d) the upper portion is of concave lens for the distant vision and lower part is of convex lens for the near vision.

14. In bifocal lenses used for the correction of presbyopia : **1**

(a) the upper portion is of convex lens for the near vision and lower part is of concave lens for the distant vision.

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(c) the upper portion is of concave lens is for the near vision and lower part is of convex lens for the distant vision.

(d) the upper portion is of concave lens for the distant vision and lower part is of convex lens for the near vision.

(d)/ The upper portion is of concave lens for the distant vision and lower part is of convex lens for the near vision.

25. (a) A person suffering from an eye defect uses lenses of power –1 D. Name the defect of vision and list its two causes. State the nature (converging/diverging) of the corrective lens. **2**

25. (a) A person suffering from an eye defect uses lenses of power –1 D. Name the defect of vision and list its two causes. State the nature (converging/diverging) of the corrective lens. **2**

(a)	
• Myopia	½
• Two causes :	
Excessive curvature of eye lens	½
Elongation of eye ball	½
• Diverging lens	½

(b) What is presbyopia ? Name the type of lenses used for the correction of this defect. State the nature (converging/diverging) of the upper part of such lenses. **2**

(b) What is presbyopia ? Name the type of lenses used for the correction of this defect. State the nature (converging/diverging) of the upper part of such lenses. **2**

(b)	
• The power of accommodation of eye lens usually decreases with ageing and the person finds it difficult to see nearby objects comfortably and distinctly.	1
• Convex lens (Bifocal lens if the person has myopia also.)	½
• The upper part of bifocal lens will be diverging.	½

19. **Assertion (A) :** Myopic eye cannot see distant objects distinctly. **1**

Reason (R) : For the correction of myopia converging lenses of appropriate power are prescribed by eye-surgeons.

19. **Assertion (A) :** Myopic eye cannot see distant objects distinctly. 1

Reason (R) : For the correction of myopia converging lenses of appropriate power are prescribed by eye-surgeons.

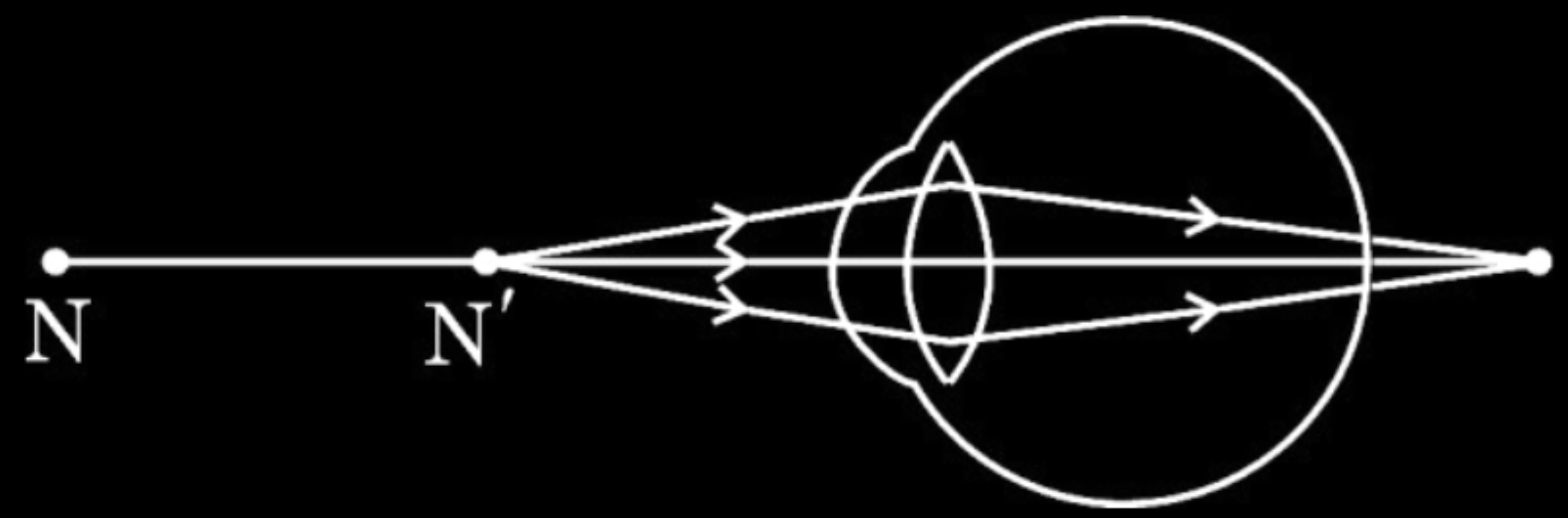
(C) / (A) is true, but (R) is false.

24. When do we say that a particular person is suffering from hypermetropia ? List two causes of this defect. Name the type of lens used to correct this defect. **2**

24. When do we say that a particular person is suffering from hypermetropia ? List two causes of this defect. Name the type of lens used to correct this defect. **2**

• When he cannot see nearby objects distinctly but can see far object clearly.	½
• **2 causes:**	
Focal length of the eye lens is too long.	½
Eyeball becomes too small.	½
• Convex or Converging lens	½

33. (a) Study the diagram given below and answer the questions that follow :



(i) Name the defect of vision depicted in this diagram stating the part of the eye responsible for this condition.

(ii) List two causes of this defect.

(iii) Name the type of lens used to correct this defect and state its role in this case. *3*

(a)	
(i) • Hypermetropia	½
• Ciliary muscles/ eye lens	½
(ii) • Focal length of the eye lens is too long.	½
• Eyeball becomes too small.	½
(iii) Converging lenses/ convex lens	½
They provide the additional focussing power required for forming the image on the retina./ Decrease the focal length of the eye lens	½

25. A person suffering from presbyopia needs bifocal lens. If he needs two lens of power – 4·0 dioptre and + 2·0 dioptre, which one of these two lenses is for the correction of distant vision and what is its focal length ? *2*

25. A person suffering from presbyopia needs bifocal lens. If he needs two lens of power – 4·0 dioptre and + 2·0 dioptre, which one of these two lenses is for the correction of distant vision and what is its focal length ? *2*

• For distant vision: lens of power – 4·0 dioptre	1
• f $(meter) = \frac{1}{P} = \frac{1}{-4{\cdot}0D} = -0.25$ m or -25 cm	1

(a) Name the type of lens presented by opticians for the correction of hypermetropia. State the role of such lenses in correcting the vision of the person suffering from this defect. 2

(a) Name the type of lens presented by opticians for the correction of hypermetropia. State the role of such lenses in correcting the vision of the person suffering from this defect. 2

(a) • Convex lens/ converging lens • It provides the additional focussing power required for forming the image on the retina. / These lenses decrease the focal length of the eye lens, thereby forming the image on the retina.	1 1

5. Consider the following statements :

i. Ciliary muscles adjust for changing the intensity of light entering the eye.

ii. Myopic eye can be corrected by converging lenses of suitable power.

iii. The function of the pupil is to regulate the quantity of light entering the eye.

iv. When ciliary muscles are completely relaxed, the focal length of the eye lens is maximum.

The correct statements are :

(A) i, ii and iv

(B) i, ii and iii

(C) iii and iv

(D) ii and iv

5. Consider the following statements :

i. Ciliary muscles adjust for changing the intensity of light entering the eye.

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(B) i, ii and iii

(C) iii and iv

(D) ii and iv

(C) /iii and iv

14. A person is advised to use lenses of power – 0·25 D for the correction of his defect of vision. It implies that the person is suffering from :

(A) hypermetropia and he has to use converging lens of focal length 25 cm in his spectacles.

(B) hypermetropia and he has to use diverging lens of focal length 4 m in his spectacles.

(C) myopia and he has to use diverging lens of focal length 4 m in his spectacles.

(D) myopia and he has to use diverging lens of focal length 25 cm in his spectacles.

14. A person is advised to use lenses of power – 0·25 D for the correction of his defect of vision. It implies that the person is suffering from :

(A) hypermetropia and he has to use converging lens of focal length 25 cm in his spectacles.

(B) hypermetropia and he has to use diverging lens of focal length 4 m in his spectacles.

(C) myopia and he has to use diverging lens of focal length 4 m in his spectacles.

(D) myopia and he has to use diverging lens of focal length 25 cm in his spectacles.

(C) myopia and he has to use diverging lens of focal length 4 m in his spectacles.

25. (a) The far point of a defective eye is nearer than infinity.

(i) Identify the defect of vision.

(ii) List two possible causes responsible for this defect.

(iii) Name the type of lens (converging/diverging) used for the correction of this defect. *2*

25. (a) The far point of a defective eye is nearer than infinity.

(i) Identify the defect of vision.

(ii) List two possible causes responsible for this defect.

(iii) Name the type of lens (converging/diverging) used for the correction of this defect. *2*

(a) (i) Myopia	½
(ii) • Excessive curvature of eye lens / Converging power of eye lens increases • Elongation of the eye ball.	½ ½
(ii) Diverging lens	½

(B) Name the type of lenses required by the persons for the correction of their defect of vision called presbyopia. Write the structure of the lenses commonly used for the correction of this defect giving reason for such designs. **2**

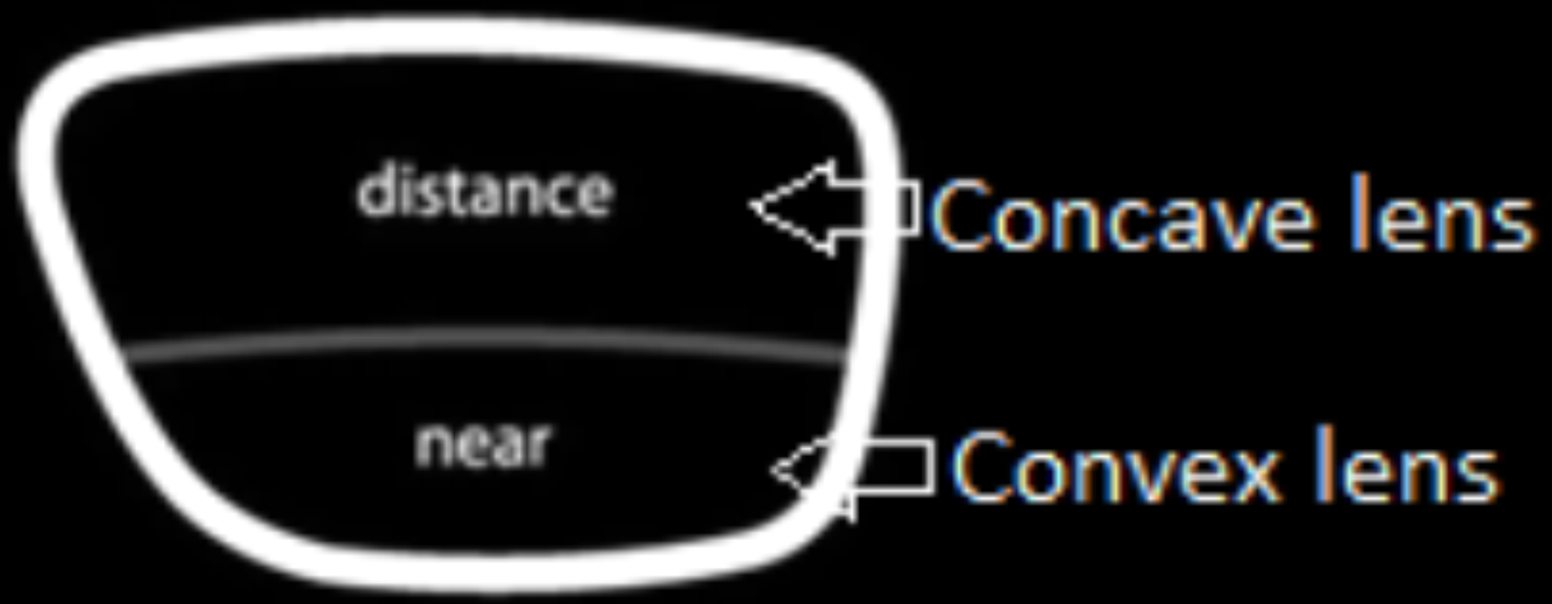
(B) Name the type of lenses required by the persons for the correction of their defect of vision called presbyopia. Write the structure of the lenses commonly used for the correction of this defect giving reason for such designs. **2**

(B)

I.

Bi-focal lens. ½

Bi-focal lens having upper portion consists of a concave lens and lower portion consists convex lens. / 1



to facilitate the distant and near vision respectively. ½

32. A person uses lenses of power –0.5 D in his spectacles for the correction of his vision.

(a) Name the defect of vision the person is suffering from.

(b) List two causes of this defect.

(c) Determine the focal length of the lenses used in the spectacles. **3**

32. A person uses lenses of power –0.5 D in his spectacles for the correction of his vision.

(a) Name the defect of vision the person is suffering from.

(b) List two causes of this defect.

(c) Determine the focal length of the lenses used in the spectacles. **3**

(a)Myopia / Near Sightedness/ Short sightedness	1
(b)Two causes:	
• Excessive curvature of eye lens	½
• Eye ball is elongated	½
(c) $f(m) = 1/P = 1/-0.5 = -2$ m	1

32. A person uses lenses of +2.0 D power in his spectacles for the correction of his vision.

(a) Name the defect of vision the person is suffering from.

(b) List two causes of this defect.

(c) Determine the focal length of the lenses used in the spectacles. **3**

32. A person uses lenses of +2.0 D power in his spectacles for the correction of his vision.

(a) Name the defect of vision the person is suffering from.

(b) List two causes of this defect.

(c) Determine the focal length of the lenses used in the spectacles. **3**

(a) Hypermetropia	1
(b) Two causes: • Eye ball has become too small /Eye ball is shortened • The focal length of the eye Lens is long / eye lens becomes less convergent	½ x2
(c) Focal length $= \frac{1}{P}$	½
$= \frac{1}{2} = +0.5$ m	½

39. The human eye is one of the most valuable and sensitive sense organs. Eyes enable us to see the wonderful world around us. We observe that some persons cannot see the objects distinctly and comfortably. The vision becomes blurred due to the refractive defects of the eye. **4**

(a) When do we say that a person is suffering from myopia ?

(b) List two main causes of myopia.

(c) (i) When do we say that a person is suffering from presbyopia. Write the structure of common type of bifocal lenses used by the persons suffering from presbyopia.

OR

(ii) A person is using corrective lenses of power +0.25 D in his spectacles for seeing the objects clearly. Name the defect of vision he is suffering from. Also calculate the focal length of the lenses used in his spectacles.

(a) When a person cannot see distant objects distinctly but can see nearby objects clearly.	1
(b) Two main reasons of myopia:	
• Excessive curvature of the eye lens	½
• Elongation of the eye ball	½
(c) (i) • The power of accommodation of the eye usually decreases with ageing, the near point gradually recedes away and find it difficult to see nearby objects comfortably and distinctly without corrective eye glasses.	1
Structure of bifocal lens :	
• Upper portion consists of concave lens.	½
• Lower portion consists of convex lens	½
OR	
(ii) Power P = + 0·25D	
• Hypermetropia	1
• Focal length = $\frac{1}{Power}$	½
$\therefore$ Focal length = $\frac{1}{+0 \cdot 25D}$ = + 4·0 m	½

14. A boy while reading a book, keeps it much beyond 25 cm from his eyes. This defect of vision has arisen because of

(A) excessive curvature of the eye lens

(B) the focal length of the eye lens has increased

(C) the eye ball has elongated

(D) the focal length of the eye lens has become too small

14. A boy while reading a book, keeps it much beyond 25 cm from his eyes. This defect of vision has arised because of

(A) excessive curvature of the eye lens

(B) the focal length of the eye lens has increased

(C) the eye ball has elongated

(D) the focal length of the eye lens has become too small

(B) / The focal length of the eye lens has increased

14. The possible way to restore clear vision of those people whose eyeball has elongated is the use of suitable

(A) bifocal lens (B) concave lens

(C) converging lens (D) convex lens

14. The possible way to restore clear vision of those people whose eyeball has elongated is the use of suitable

(A) bifocal lens
(B) concave lens
(C) converging lens
(D) convex lens

(B) / Concave lens

31. A person is suffering from an eye defect in which the far point of the eye is much nearer than infinity. Name the defect of vision the person is suffering from. List two main causes of this defect. Write the type of the corrective lens and draw a ray diagram to show the function of the corrective lens. *3*

31. A person is suffering from an eye defect in which the far point of the eye is much nearer than infinity. Name the defect of vision the person is suffering from. List two main causes of this defect. Write the type of the corrective lens and draw a ray diagram to show the function of the corrective lens. *3*

• Myopia /Shortsightedness	½
Causes :	
(i) Excessive curvature of the eye lens	½
(ii) Elongation of the eye ball	½
• Diverging lens/concave lens	½
• 	1

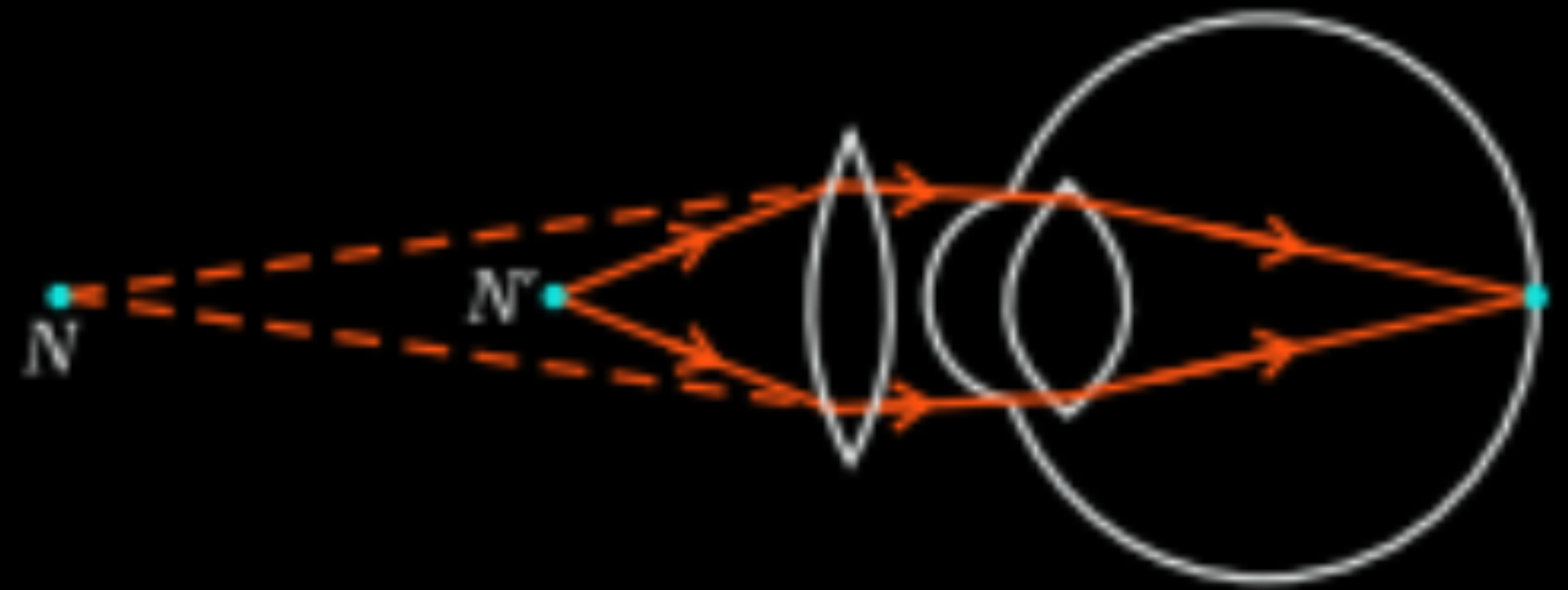
25. (a) A student has difficulty in reading his textbooks but can read the blackboard clearly while sitting in the last row. Name the defect of vision the student is suffering from. List two reasons due to which this defect arises. Write the nature of the lenses required to correct this defect. *2*

25. (a) A student has difficulty in reading his textbooks but can read the blackboard clearly while sitting in the last row. Name the defect of vision the student is suffering from. List two reasons due to which this defect arises. Write the nature of the lenses required to correct this defect. *2*

(a) Hypermetropia /Farsightedness/Longsightedness.	½
Reasons:	
(i) Focal length of the eye lens is too long	½
(ii) Eyeball becomes too small.	½
Correction	
Convex lens /Converging lens	½

31. A person has to keep reading material much beyond 25 cm (say at 50 cm) from the eye for comfortable reading. Name the defect of vision he is suffering from. List two causes responsible for arising of this defect. Draw a labelled diagram showing correction of this defect using eye-glasses. Are these glasses convergent or divergent of light ? *3*

31. A person has to keep reading material much beyond 25 cm (say at 50 cm) from the eye for comfortable reading. Name the defect of vision he is suffering from. List two causes responsible for arising of this defect. Draw a labelled diagram showing correction of this defect using eye-glasses. Are these glasses convergent or divergent of light ? *3*

• Hypermetropia/ far-sightedness	½
• (i) Focal length of the eye lens is too long (ii) Eye ball has become too small/shortened	½x2
	1
• Convergent of light	½

34. (a) (i) The power of a lens 'X' is – 2·5 D. Name the lens and determine its focal length in cm. For which eye defect of vision will an optician prescribe this type of lens as a corrective lens ?

34. (a) (i) The power of a lens 'X' is – 2·5 D. Name the lens and determine its focal length in cm. For which eye defect of vision will an optician prescribe this type of lens as a corrective lens ?

(a) (i)	
• Concave lens	½
• $P = \frac{1}{f(m)}$	½
$-2 \cdot 5 = \frac{1}{f}$	
$f = \frac{10}{-2 \cdot 5} = -0 \cdot 4 \text{ m} = -40 \text{ cm}$	½
• Myopia	½

14. An old person is suffering from an eye defect caused by weakening of ciliary muscles and diminishing flexibility of the eye lens. If the defect of vision is 'a' which can be corrected by lens 'b', then 'a' and 'b' respectively are :

(A) hypermetropia and convex lens

(B) presbyopia and bifocal lens

(C) myopia and concave lens

(D) myopia and bifocal lens

14. An old person is suffering from an eye defect caused by weakening of ciliary muscles and diminishing flexibility of the eye lens. If the defect of vision is ‘a’ which can be corrected by lens ‘b’, then ‘a’ and ‘b’ respectively are :

(A) hypermetropia and convex lens

(B) presbyopia and bifocal lens

(C) myopia and concave lens

(D) myopia and bifocal lens

B / Presbyopia and bifocal lens

24. A student uses spectacles of focal length – 2·5 m.

(a) Name the defect of vision he is suffering from.

(b) Which lens is used for the correction of this defect ?

(c) List two main causes of developing this defect.

(d) Compute the power of this lens. *3*

24. A student uses spectacles of focal length – 2·5 m.

(a) Name the defect of vision he is suffering from.

(b) Which lens is used for the correction of this defect ?

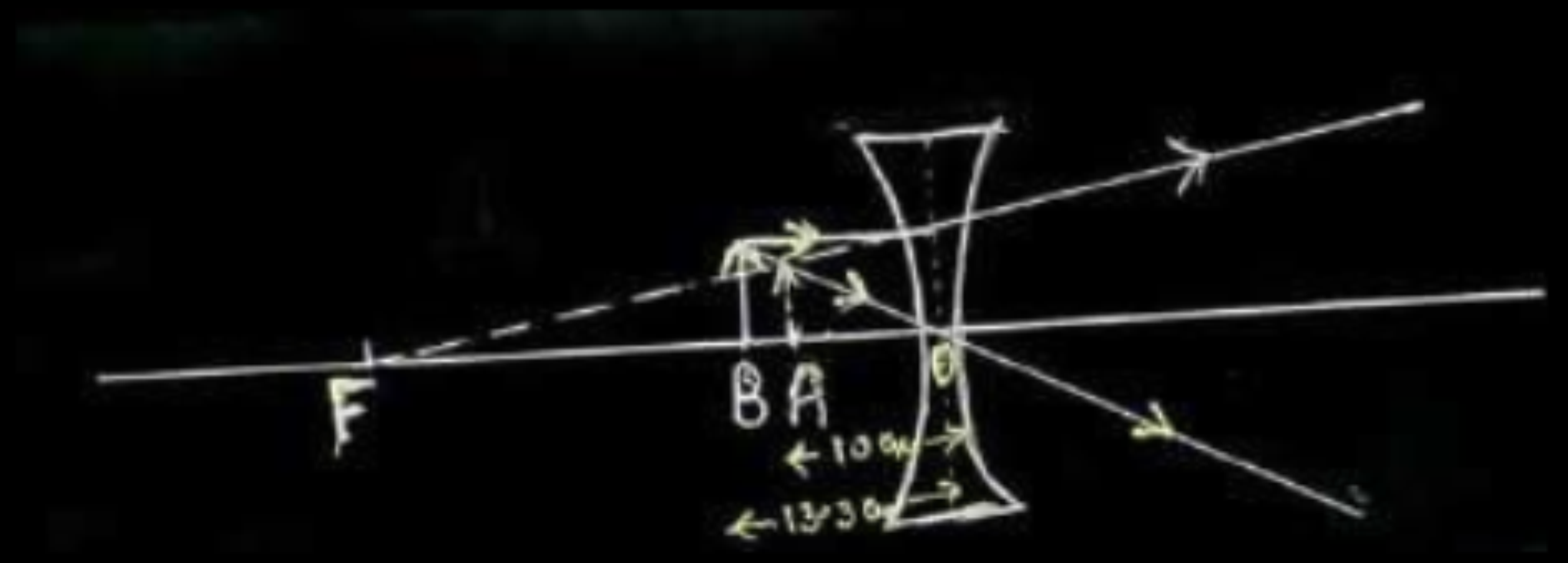
(c) List two main causes of developing this defect.

(d) Compute the power of this lens. *3*

(a) Myopia/Short sightedness	½
(b) Concave/Diverging lens.	½
(c) • Excessive curvature of eye lens • elongation of eye ball	½+ ½
(d) $P(D) = \frac{1}{f(m)}$ $P(D) = \frac{1}{-2.5(m)} = \frac{10}{-25} = \frac{2}{-5} = -0.4D$ (Deduct ½ mark if unit is not mentioned)	1

(a) A person suffering from myopia (near-sightedness) was advised to wear corrective lens of power – 2·5 D. A spherical lens of same focal length was taken in the laboratory. At what distance should a student place an object from this lens so that it forms an image at a distance of 10 cm from the lens ?

(b) Draw a ray diagram to show the position and nature of the image formed in the above case. *5*

(a) $f(m) = \frac{1}{P(D)}$	½
$$f = \frac{1}{-2.5D} = \frac{-10}{25D} = -0.4m = -40cm$$	1
f=-40 cm v =-10 cm u = ?	
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$	½
$$\frac{1}{-10\ cm} - \frac{1}{u} = \frac{1}{-40\ cm}$$	
$$-\frac{1}{u} = \frac{1}{-40} + \frac{1}{10}$$	½
$$= \frac{-1+4}{40} = \frac{3}{40}$$	
$\therefore u = -\frac{40}{3} = -13.3$ cm	1
(b) Since the power is –ve , the lens used is concave / diverging	½
 	1
OA = v = -10cm ; OB = u =-13.3 cm ; OF = f = -40cm	

23. A person may suffer from both myopia and hypermetropia defects.

(a) What is this condition called ?

(b) When does it happen ?

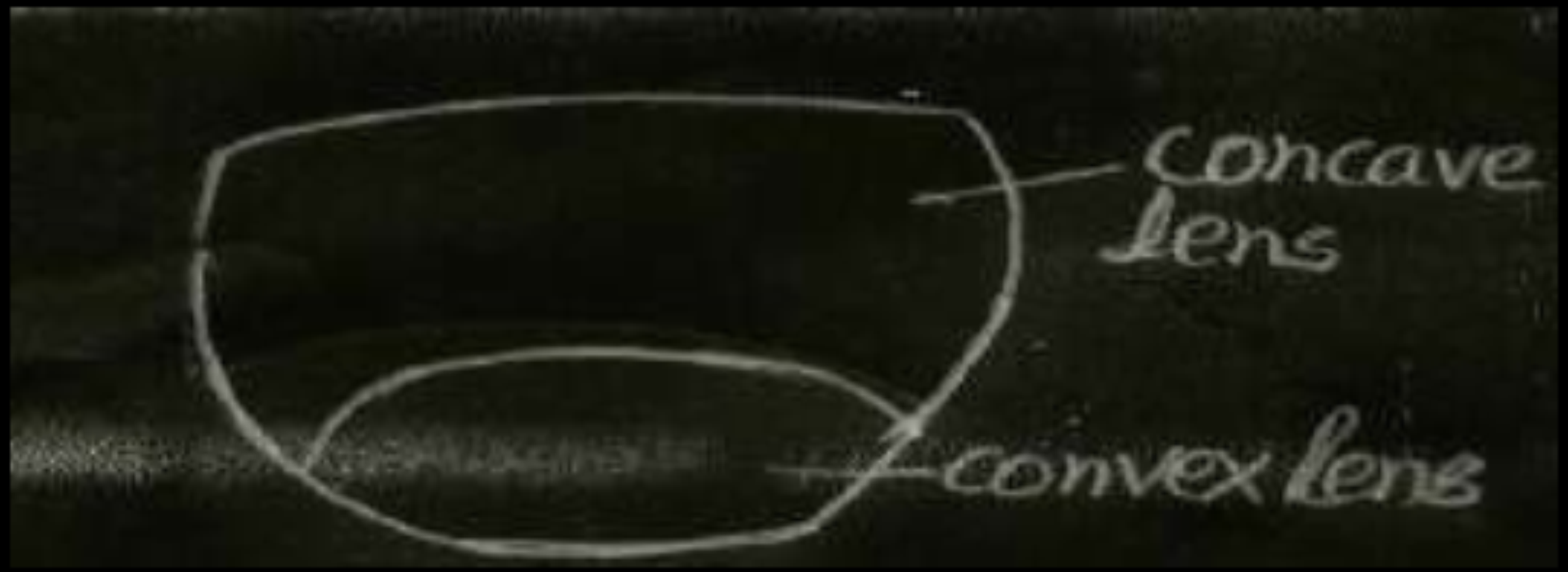
(c) Name the type of lens often required by the persons suffering from this defect. Draw labelled diagram of such lenses. *3*

23. A person may suffer from both myopia and hypermetropia defects.

(a) What is this condition called ?

(b) When does it happen ?

(c) Name the type of lens often required by the persons suffering from this defect. Draw labelled diagram of such lenses. *3*

(a) Presbyopia	½
(b) Gradual weakening of the ciliary muscles of the eye/ diminishing flexibility of the eye lens.	1
(c) Bifocal lens	½
 	1

(a) A person is suffering from both myopia and hypermetropia.

(i) What kind of lenses can correct this defect ?

(ii) How are these lenses prepared ?

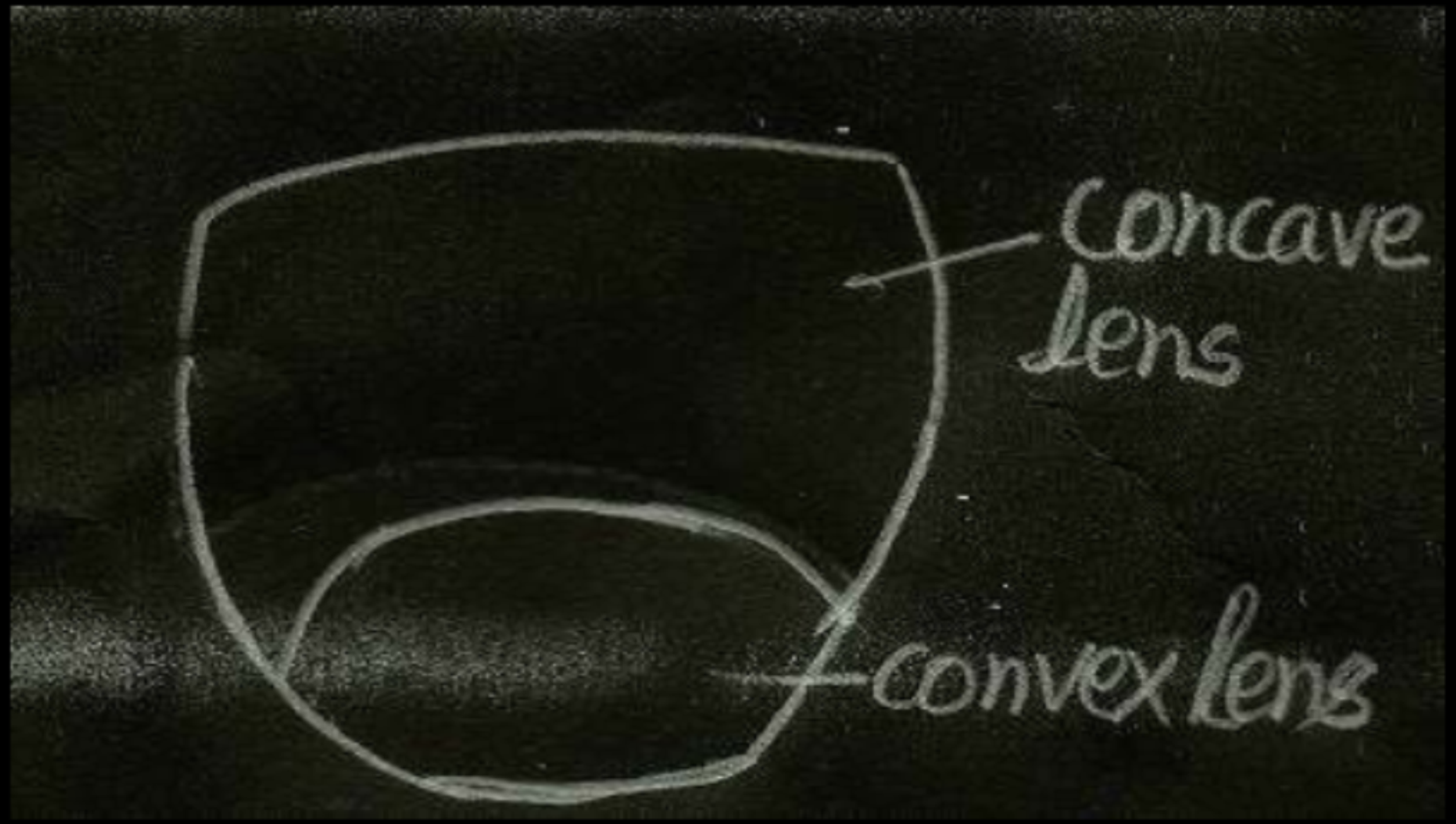
(b) A person needs a lens of power + 3D for correcting his near vision and –3D for correcting his distant vision. Calculate the focal lengths of the lenses required to correct these defects. **3**

(a) A person is suffering from both myopia and hypermetropia.

(i) What kind of lenses can correct this defect ?

(ii) How are these lenses prepared ?

(b) A person needs a lens of power + 3D for correcting his near vision and –3D for correcting his distant vision. Calculate the focal lengths of the lenses required to correct these defects. **3**

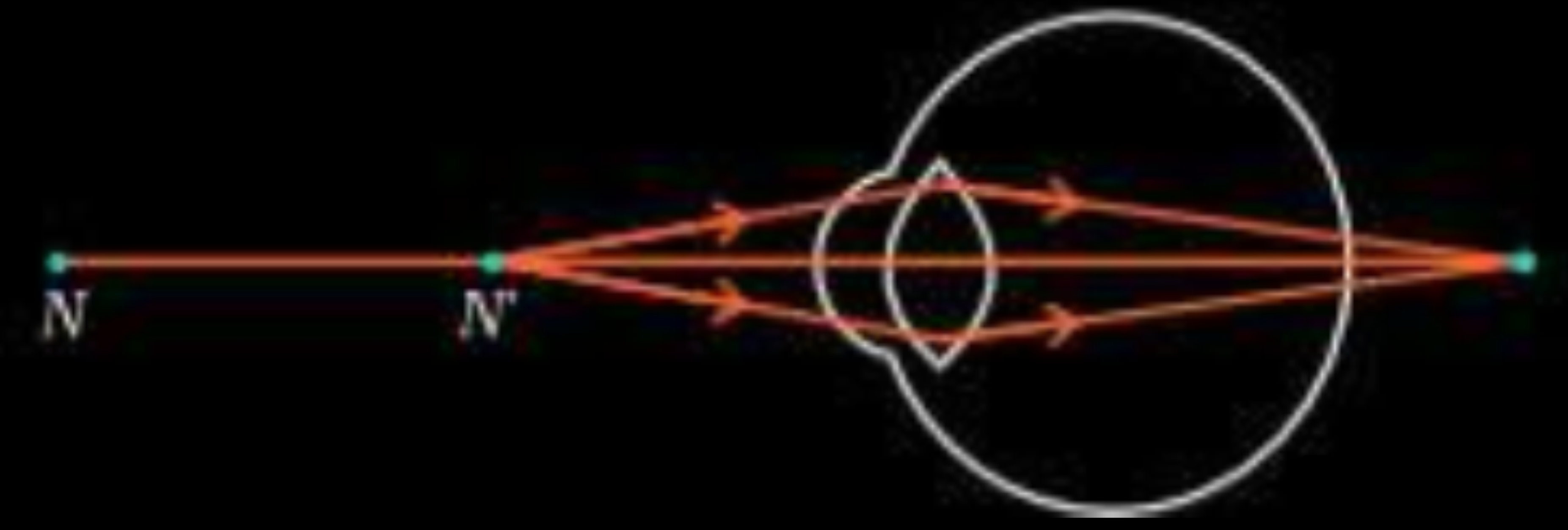
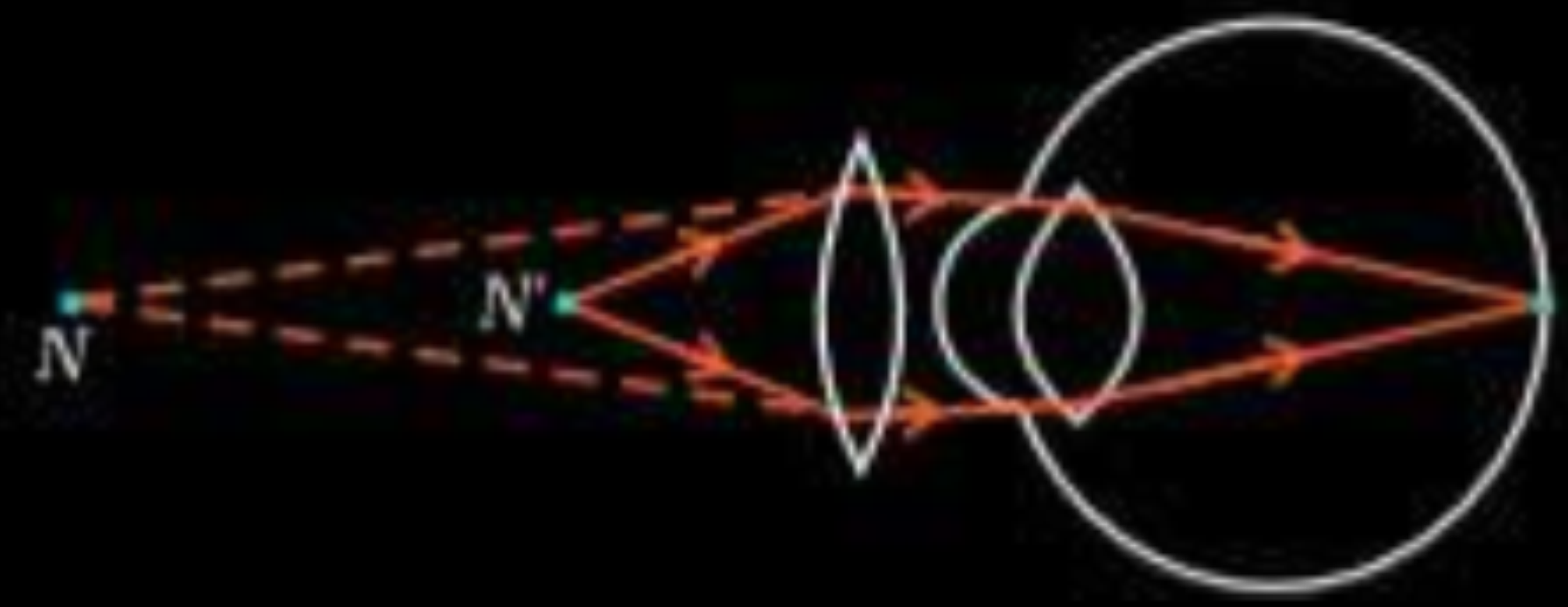
(a) (i) Bifocal Lens	½
(ii) Upper part of lens is concave and lower part of the lens is convex. /	½, ½
	
(b) P = +3D	
$f = \frac{1}{P}$	½
$= \frac{1}{3}\ m = \frac{+100}{3}\ cm = +\ 33.3\ cm$	½
P = - 3D	
$f = \frac{-100}{3} = \ -33.3\ cm$	½

24. (a) List two causes of hypermetropia.

(b) Draw ray diagrams showing (i) a hypermetropic eye and (ii) its correction using suitable optical device. **3**

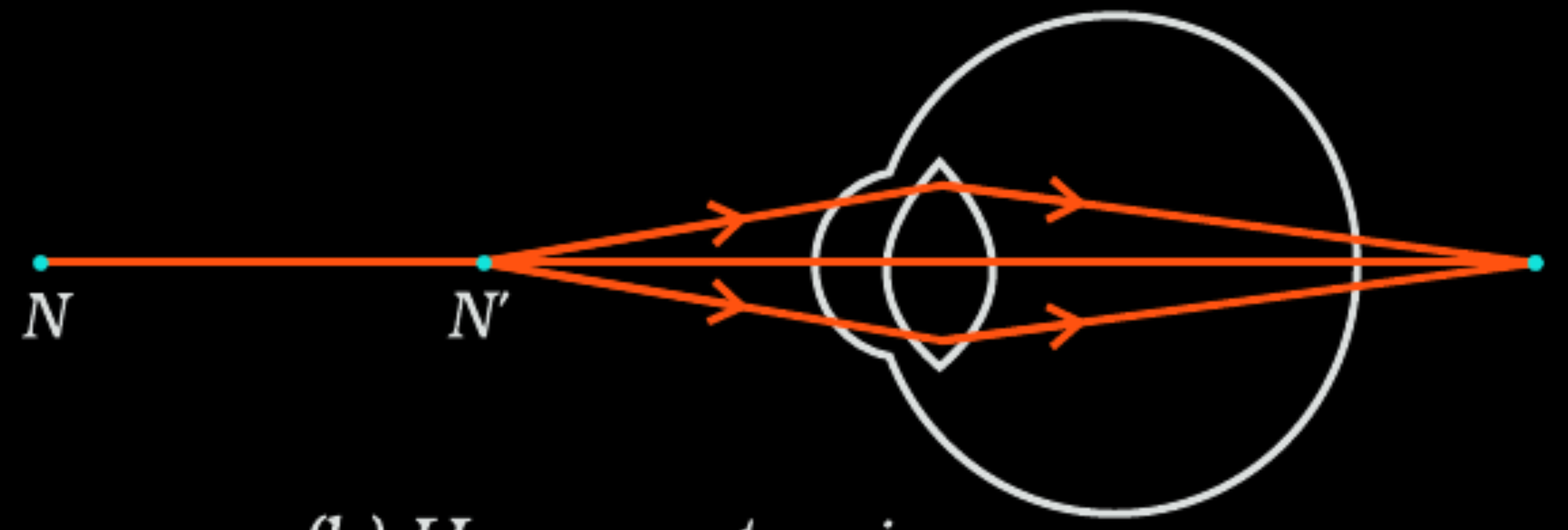
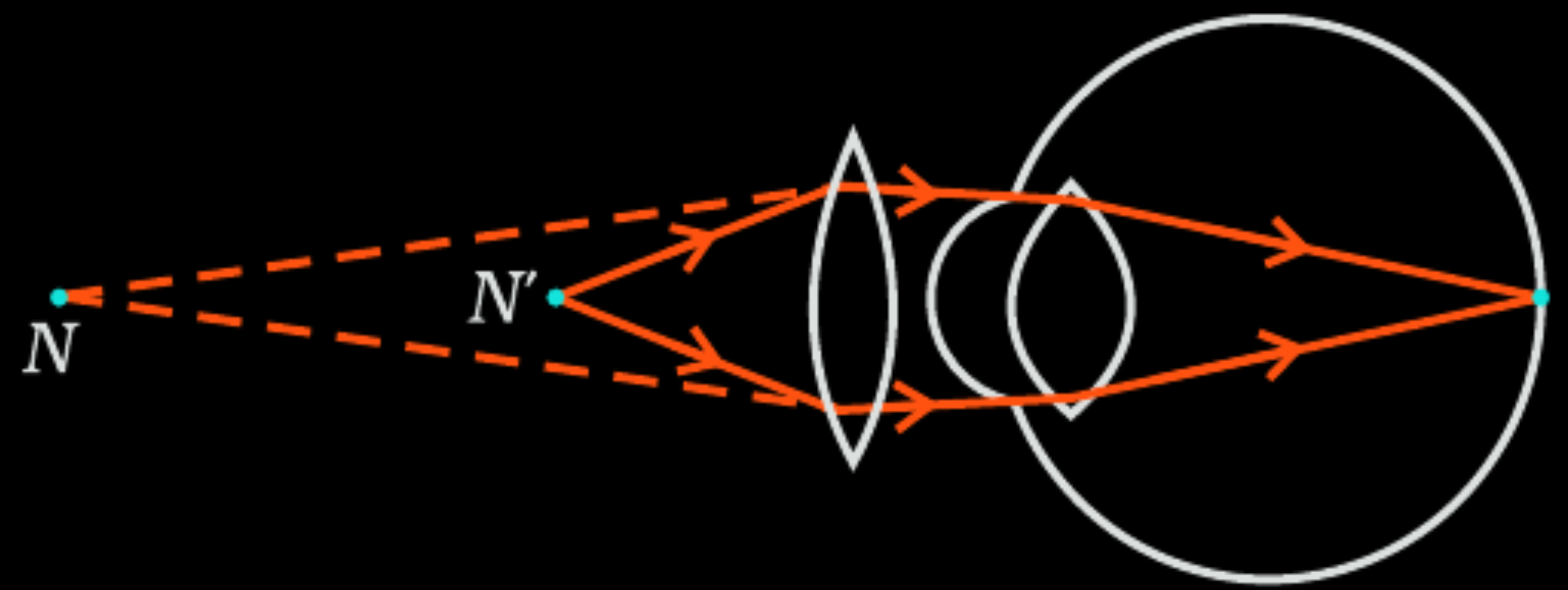
24. (a) List two causes of hypermetropia.

(b) Draw ray diagrams showing (i) a hypermetropic eye and (ii) its correction using suitable optical device. **3**

(a) (i) Size of eyeball decreases (ii) Focal length of eye lens is too long / Power of eye lens decreases.	½ + ½
(b) Diagrams :  Hypermetropic Eye	1
 Corrected Eye	1

When do we consider a person to be myopic or hypermetropic ? List two causes of hypermetropia. Explain using ray diagrams how the defect associated with hypermetropic eye can be corrected.

When do we consider a person to be myopic or hypermetropic ? List two causes of hypermetropia. Explain using ray diagrams how the defect associated with hypermetropic eye can be corrected.

Myopia:- Difficult to see the objects placed far away / Hypermtropia: Difficult to see very close or nearby objects.	1
Causes of hypermetropia – (i) The focal length of the eye lens is too long (ii) eye ball has become too small	½+½
 (b) Hypermetropic eye  (c) Correction for Hypermetropic eye	
Note: Diagram with brief description -03; only correct diagram with labelling -2 or only explanation 01	3

A student needs spectacles of power – 0·5 D for the correction of his vision.

(i) Name the defect in vision the student is suffering from.

(ii) Find the nature and focal length of the corrective lens.

(iii) List two causes of this defect.

A student needs spectacles of power – 0·5 D for the correction of his vision.

(i) Name the defect in vision the student is suffering from.

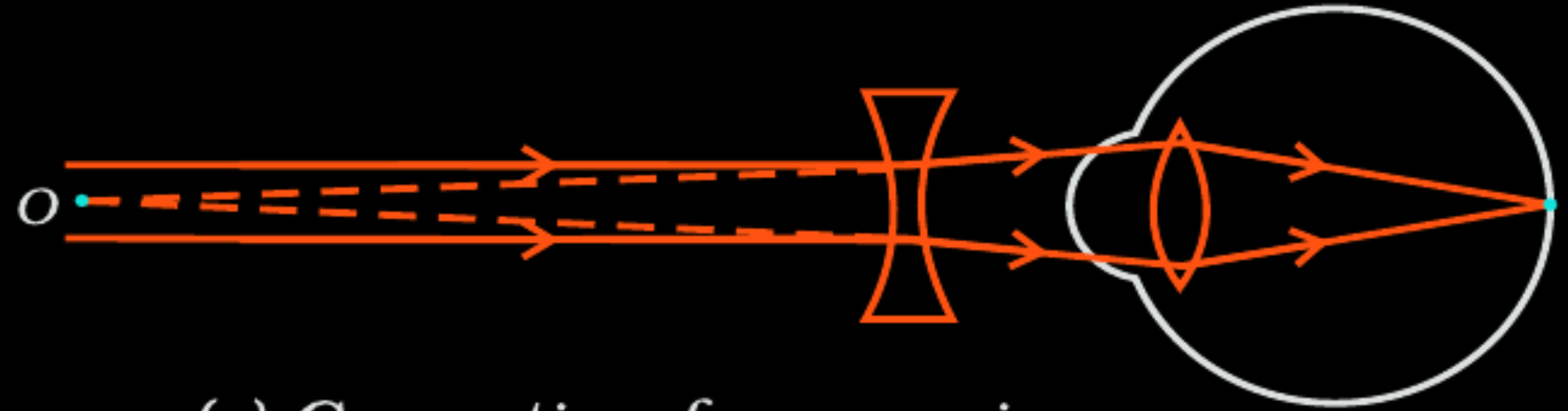
(ii) Find the nature and focal length of the corrective lens.

(iii) List two causes of this defect.

(i)	Myopia	1
(ii)	Concave / diverging lens and focal Length = 200cm	1
(iii)	(a) excessive curvature of the eye lens	½
	(b) elongation of eye ball	½

When do we consider a student sitting in the class to be myopic ? List two causes of this defect. Explain using a ray diagram how this defect of eye can be corrected.

When do we consider a student sitting in the class to be myopic ? List two causes of this defect. Explain using a ray diagram how this defect of eye can be corrected.

(i) If the student cannot see the words written on the black board then he is considered myopic.	1
(ii) The defect may arise due to 1) Excessive curvature of the eyeball 2) Elongation of the eyeball	½ x 2
(iii)  (c) Correction for myopia	1

List two causes of presbyopia. Draw labelled diagram of a lens used for the correction of this defect of vision.

List two causes of presbyopia. Draw labelled diagram of a lens used for the correction of this defect of vision.

Causes i) gradual weakening of the ciliary muscles ii) diminishing flexibility of the eye lens. (No diagram given in t.b. or reference books. Diagram for lens used, use a bifocal lens)	 1 1

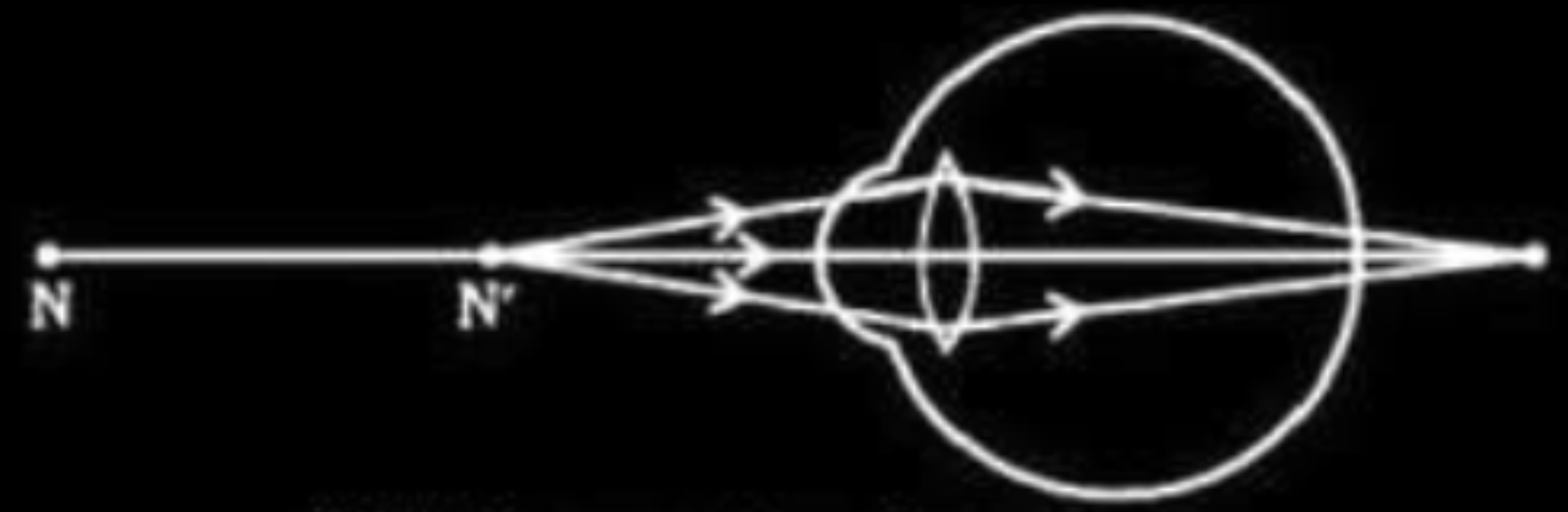
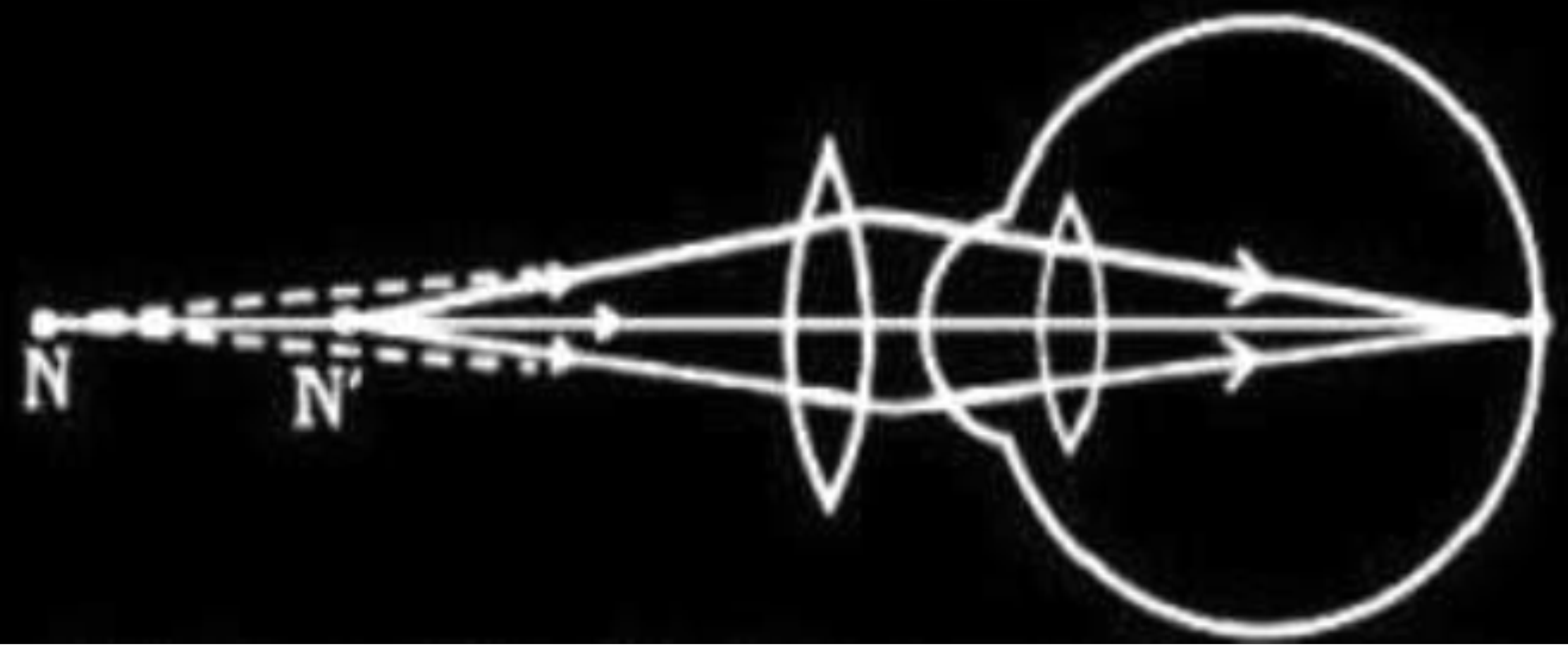
A person is unable to see objects distinctly placed within 50 cm from his eyes.

(a) Name the defect of vision the person is suffering from and list its two possible causes.

(b) Draw a ray diagram to show the defect in the above case.

(c) Mention the type of lens used by him for the correction of the defect and calculate its power. Assume that the near point for the normal eye is 25 cm.

(d) Draw a labelled diagram for the correction of the defect in the above case.

(a) Hypermetropia / farsightedness	½
Causes:	½
i. Shortening of eyeball	½
ii. Curvature of eye lens decreases / focal length of eye lens increases.	
b)  *(b) Hypermetropic eye*	1
(c) Convex lens	½+ ½
1/f=1/v-1/u	
=1/(-50cm) -1/(-25cm)	
=1/50cm	
Hence ,f=50 cm=0.5m	
There fore power =(1/0.5)D=2D	½
(d)Correction of Hypermetropia	
 	½

(a) A student is unable to see clearly the words written on the black board placed at a distance of approximately 3 m from him. Name the defect of vision the boy is suffering from. State the possible causes of this defect and explain the method of correcting it.

(b) Why do stars twinkle ? Explain.

(a) A student is unable to see clearly the words written on the black board placed at a distance of approximately 3 m from him. Name the defect of vision the boy is suffering from. State the possible causes of this defect and explain the method of correcting it.

(b) Why do stars twinkle ? Explain.

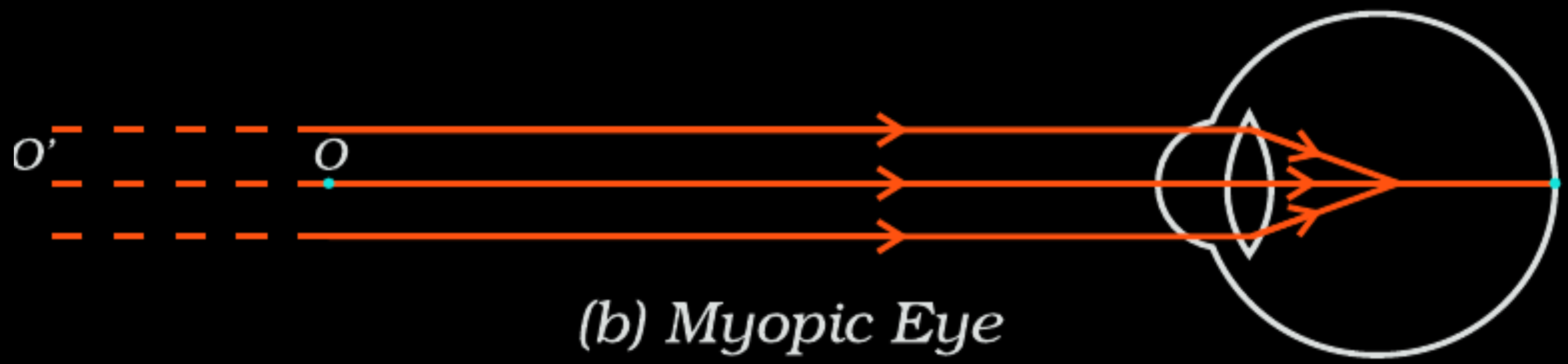
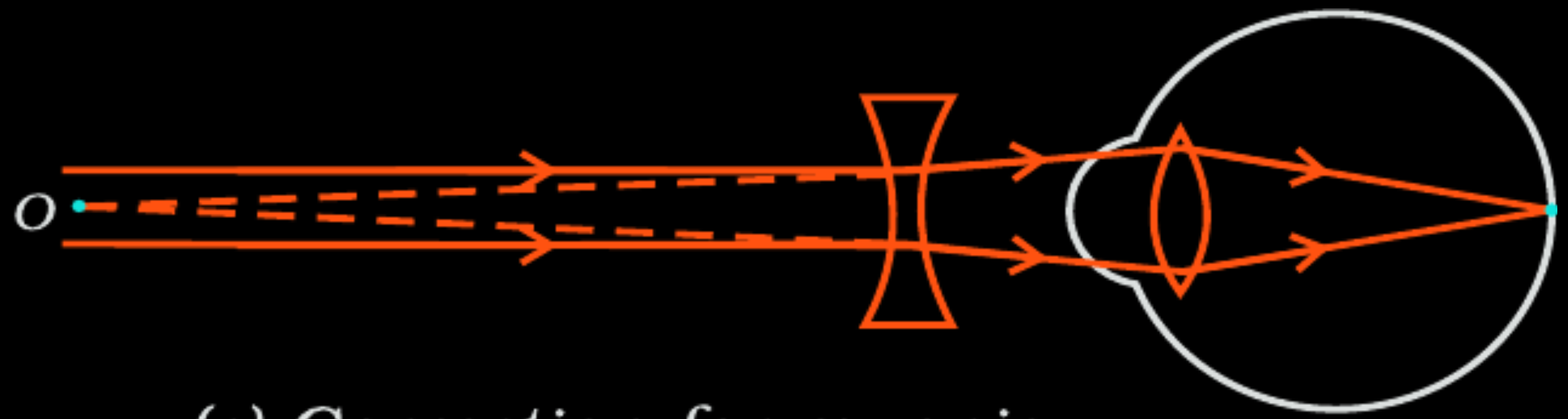
a.	<u>Defect of vision</u> – Myopia or short sightedness or near sightedness	1
	Causes of myopia: i) Excessive curvature of eye lens/eye lens becomes more converging ii) Elongation of eye ball	½ + ½
	Methods of correction: By the use of concave lens of suitable power or focal length the defect is corrected. / suitable diagrammatic representation.	1
b.	<u>Due to atmospheric refraction</u>	1
	The density of different layers of air keeps on changing due to which the apparent image of the stars keeps on changing. This changing position of stars appears as twinkling of stars.	1

(a) A student suffering from myopia is not able to see distinctly the objects placed beyond 5 m. List two possible reasons due to which this defect of vision may have arisen. With the help of ray diagrams, explain

(i) why the student is unable to see distinctly the objects placed beyond 5 m from his eyes.

(ii) the type of the corrective lens used to restore proper vision and how this defect is corrected by the use of this lens.

(b) If, in this case, the numerical value of the focal length of the corrective lens is 5 m, find the power of the lens as per the new Cartesian sign convention.

a)	
• excessive curvature of the eye lens	½
• elongation of the eyeball	½
i)  (b) Myopic Eye	1
ii) Concave / diverging lens	½
 (c) Correction for myopia	1
b) f = — 5 m (since lens is concave)	½
$P = \frac{1}{f(\text{metre})}$	½
P= — 0.2 D	½

A student is unable to see clearly the words written on the blackboard placed at a distance of approximately 4 m from him. Name the defect of vision the boy is suffering from. Explain the method of correcting this defect. Draw ray diagram for the :

(i) defect of vision and also

(ii) for its correction.

Write the importance of ciliary muscles in the human eye. Name the defect of vision that arises due to gradual weakening of the ciliary muscles in old age. What type of lenses are required by the persons suffering from this defect to see the objects clearly ?

Akshay, sitting in the last row in his class, could not see clearly the words written on the blackboard. When the teacher noticed it, he announced if any student sitting in the front row could volunteer to exchange his seat with Akshay. Salman immediately agreed to exchange his seat with Akshay. He could now see the words written on the blackboard clearly. The teacher thought it fit to send the message to Akshay's parents advising them to get his eyesight checked.

In the context of the above event, answer the following questions :

(a) Which defect of vision is Akshay suffering from ? Which type of lens is used to correct this defect ?

• Ciliary muscles modify the curvature of the eye lens to enable the eye to focus objects at varying distances/ help in adjusting the focal length of the eye lens	1
• Presbyopia	½
• Bifocal lens	½
(a) Defect – Myopia/ Nearsightedness	½
Corrective lens – Concave/ Diverging lens	½